Portofolio

Flacintya Previar Hardiana Putri

Geodetic Engineer

ABOUT ME

Flacintya Previar Hardiana Putri

I am a Geodetic Engineering graduate from Universitas Gadjah Mada, specializing in spatial data acquisition, post-processing, and 3D digital reconstruction. My technical experience ranges from conducting mining surveys, utilizing USV and aerial photogrammetry, to developing advanced 3D architectural models (LoD 4) and interactive digital twins in Unity and Unreal Engine. Alongside my engineering background, I have directed visual branding and managed creative content production for various organizational programs.

EDUCATION

Universitas Gadjah Mada (UGM) | Bachelor of Geodetic Engineering (Aug 2022 – Jul 2026) **GPA: 3.70 / 4.00** (Cum Laude)

Thesis: Comparative analysis of RTK positioning precision and performance utilizing CORS-IGS and INACORS networks.

EXPERIENCES

- 3D Modeler (Museum Kartini Jepara) | Tim GEOLIVE UGM
- Mine Survey Engineer Intern | PT Pamapersada Nusantara MTBU
- 3D Modeler (Old Batavia Project) | Teknik Geodesi UGM
- Applied GNSS Practicum Assistant | Teknik Geodesi UGM
- Engineering Intern | PT Solusi Energy Nusantara (SENA)
- Discrete Mathematics Teaching Assistant | Teknik Geodesi UGM

Geospatial & Surveying

Creative Tools & Design

3D Modelling & Visualisation

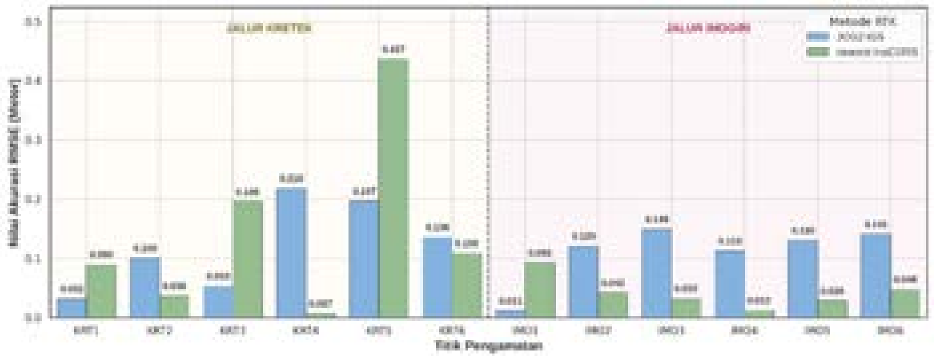
Photogrammetry, Lidar, Remote Sensing

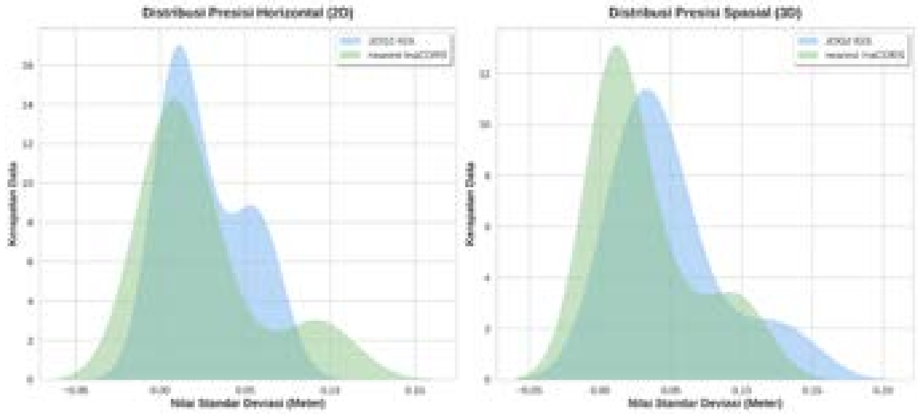
CREATIVE & LEADERSHIP EXPERIENCES

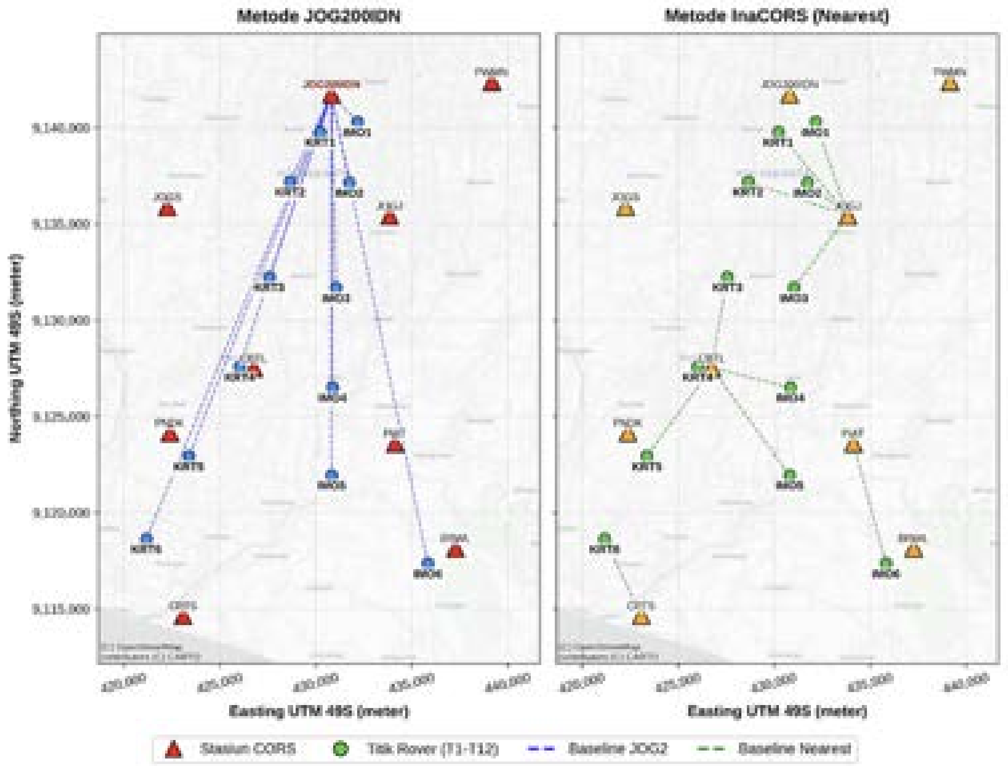
- Design & Documentation Coordinator | KKN PPM UGM Siruntu Selayar
- Creative Content Staff (Entrepreneurship Dept) | KMTG UGM
- Design & Documentation Coordinator | GEOID UGM
- Decoration Sub-Coordinator | Mercator KMTG

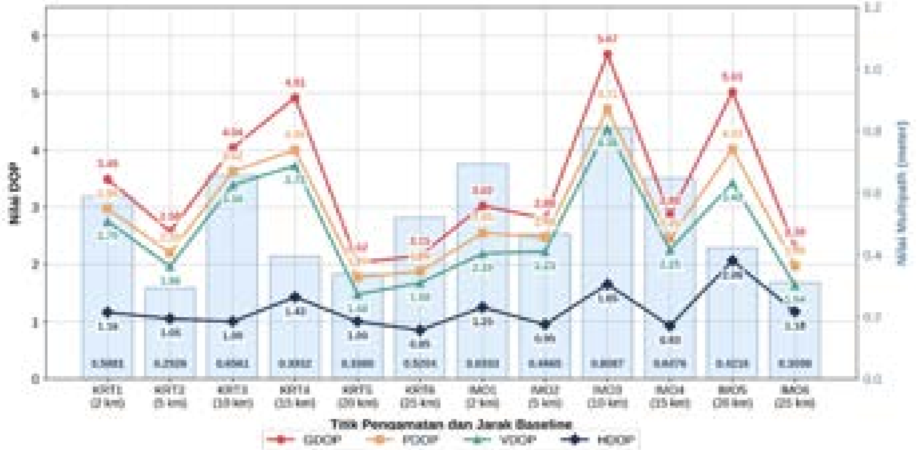
UNDERGRADUATE **THESIS**

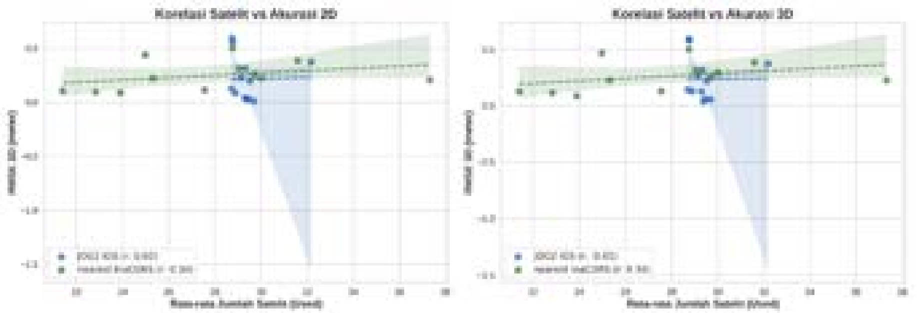
Comparative Analysis of RTK Positioning Accuracy and Performance Using JOG200IDN Single Base and Nearest Ina-CORS Services

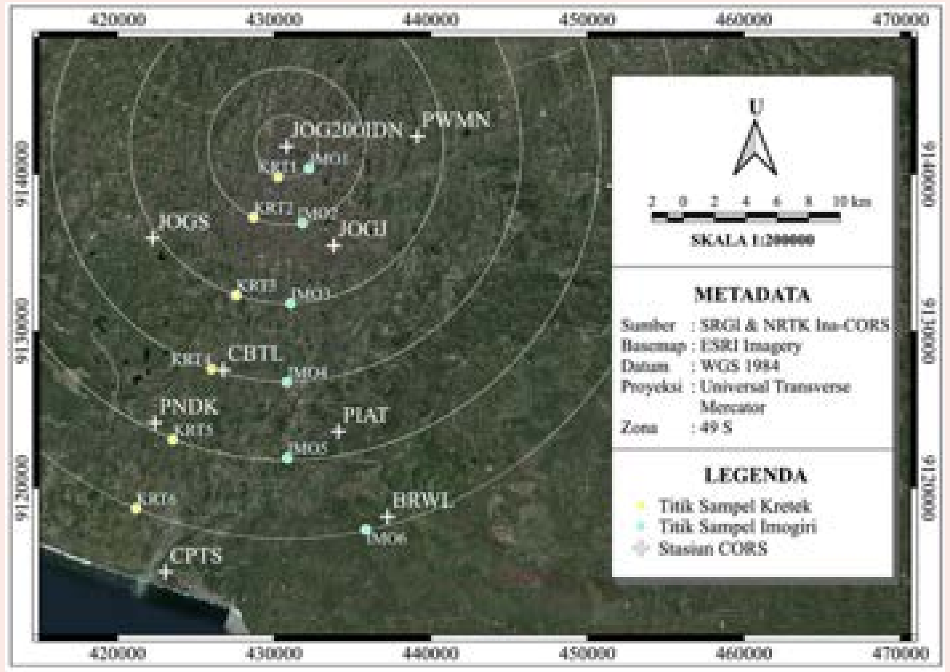












Research Objective

Evaluated the external reliability (RMSE) and internal consistency (Stdev) of global IGS standards against national dense networks across 12 observation points (2–25 km baselines) in D.I. Yogyakarta.

Analytical Methodology

Conducted rigorous spatial analysis using Static GNSS observations as ground truth (40 mins). Validated data streams using Pearson correlation, paired t-tests, and F-tests to assess multi-constellation geometry and Age of Differential (AoD).

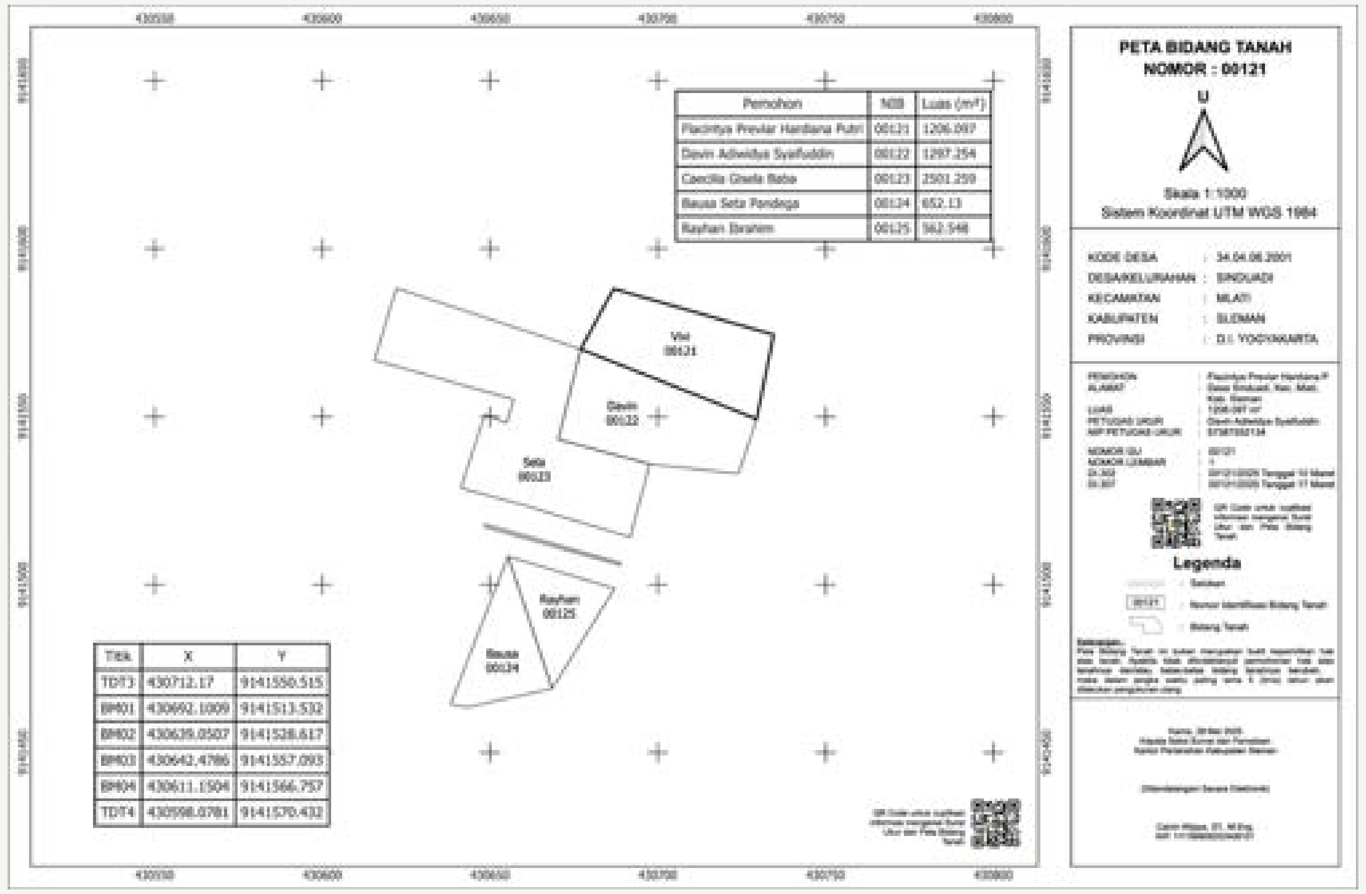
Key Impact & Findings

Proved that Ina-CORS's network density drastically mitigates latency (1.87s vs 10.14s), yielding a 100% fixed solution rate. The study provides empirical data for optimizing RTK operational thresholds and mitigating spatial decorrelation in the field.

ACADEMIC **PROJECTS**

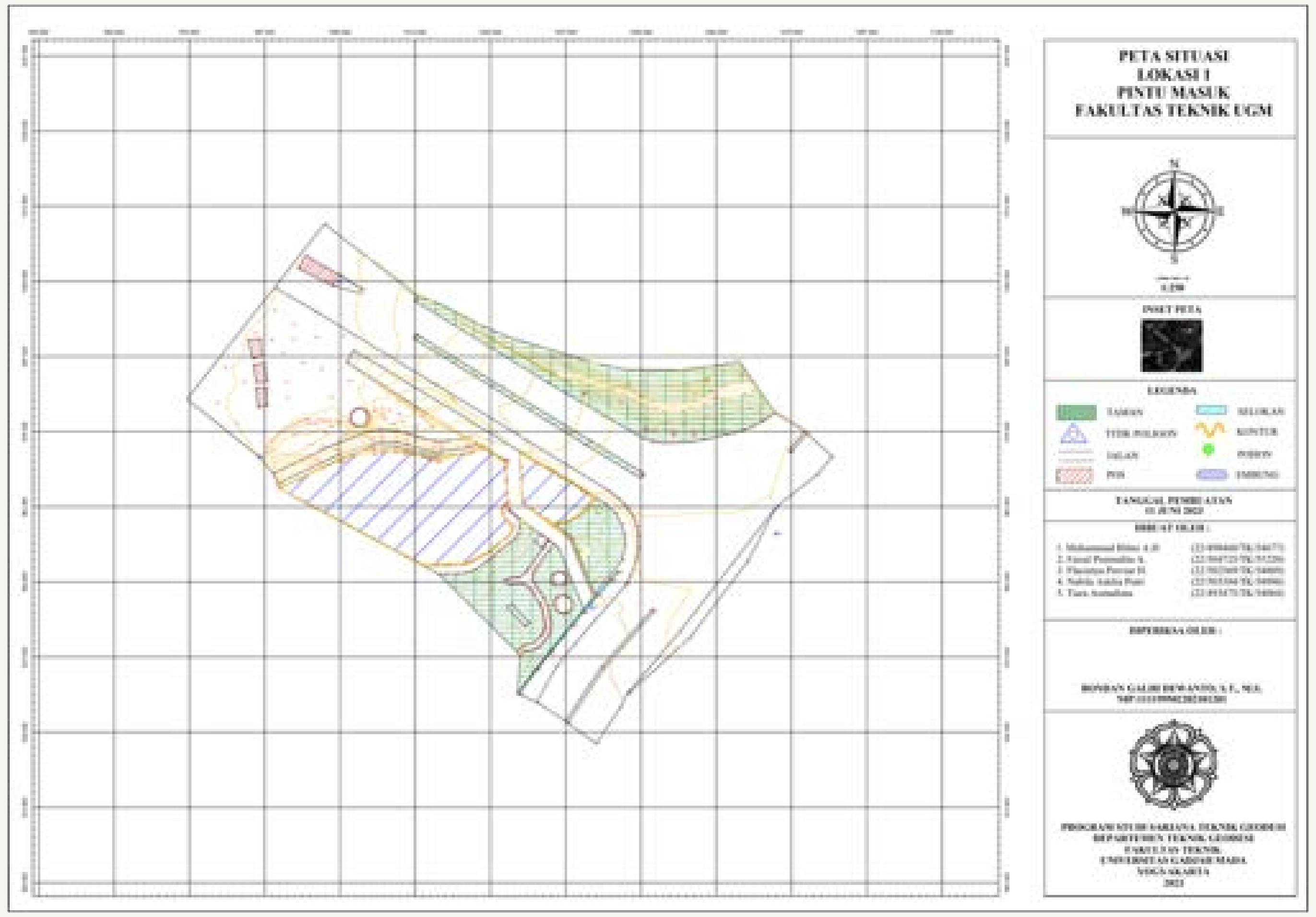
Terestrial and Cadastral Survey

These projects highlight my ability to collaborate effectively in executing topographic surveys, cadastral mapping, and geometric infrastructure design. Working effectively as a team, we successfully collected and processed precise spatial data into accurate engineering solutions



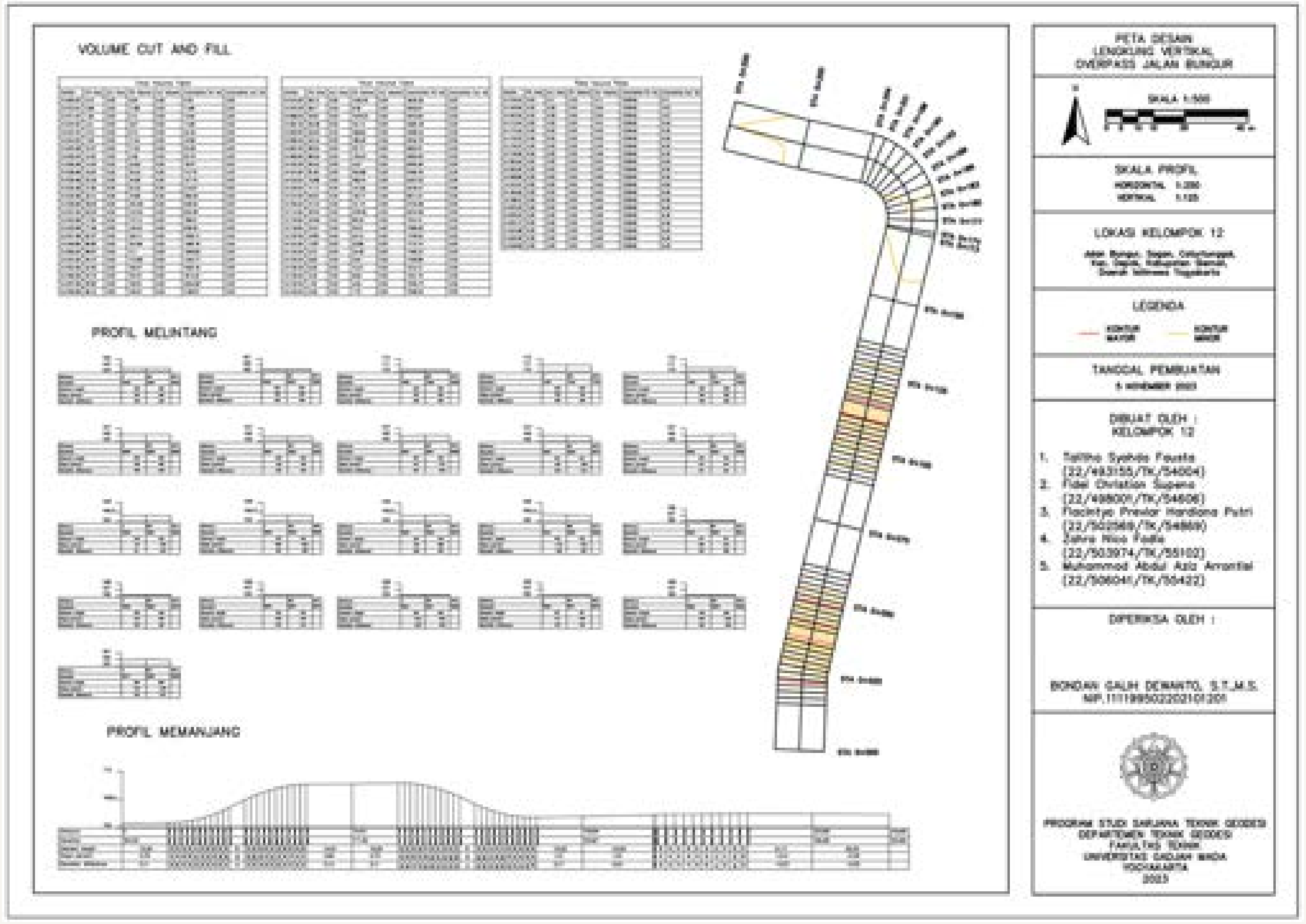
Digital Cadastral Map – Cadastral Survey

Generated an electronic cadastral map (Peta Bidang Tanah) defining precise land parcel boundaries and land areas in Sinduadi, Sleman.



Topographic Map – Terestrial Survey

Produced a topographic situation map at a 1:250 scale for the main entrance area of the UGM Faculty of Engineering.



Vertical Curve & Overpass – Terestrial Survey

Designed geometric road alignments & overpass, integrating vertical profiles and volume calculations.

ACADEMIC **PROJECTS**

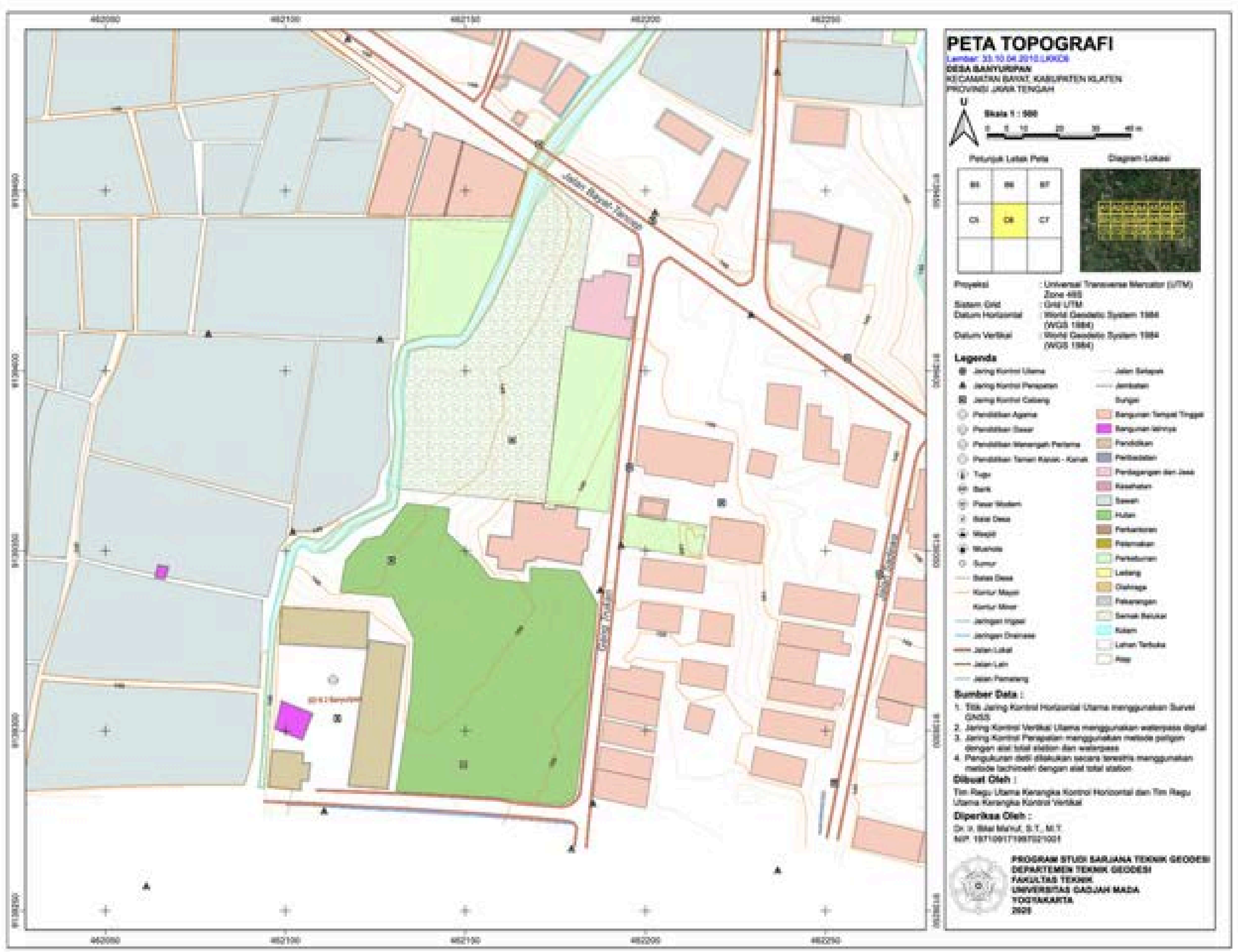
Topographic Surveying, GNSS & Cartography

Topographic Map – Field Camp

Produced a 1:500 scale topographic map in Banyuripan, Bayat, by executing comprehensive terrestrial surveying utilizing Total Stations and automatic levels to establish precise spatial controls.

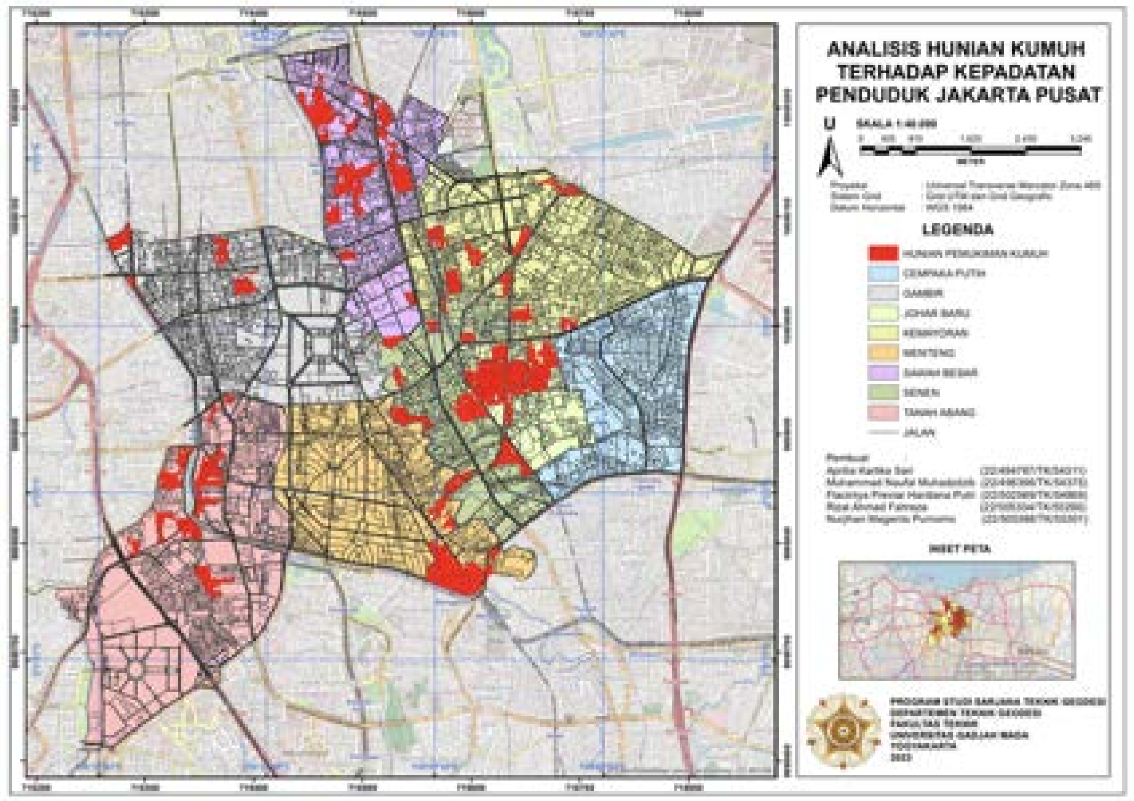
Applied GNSS Practicum – RTK Radio Detailing

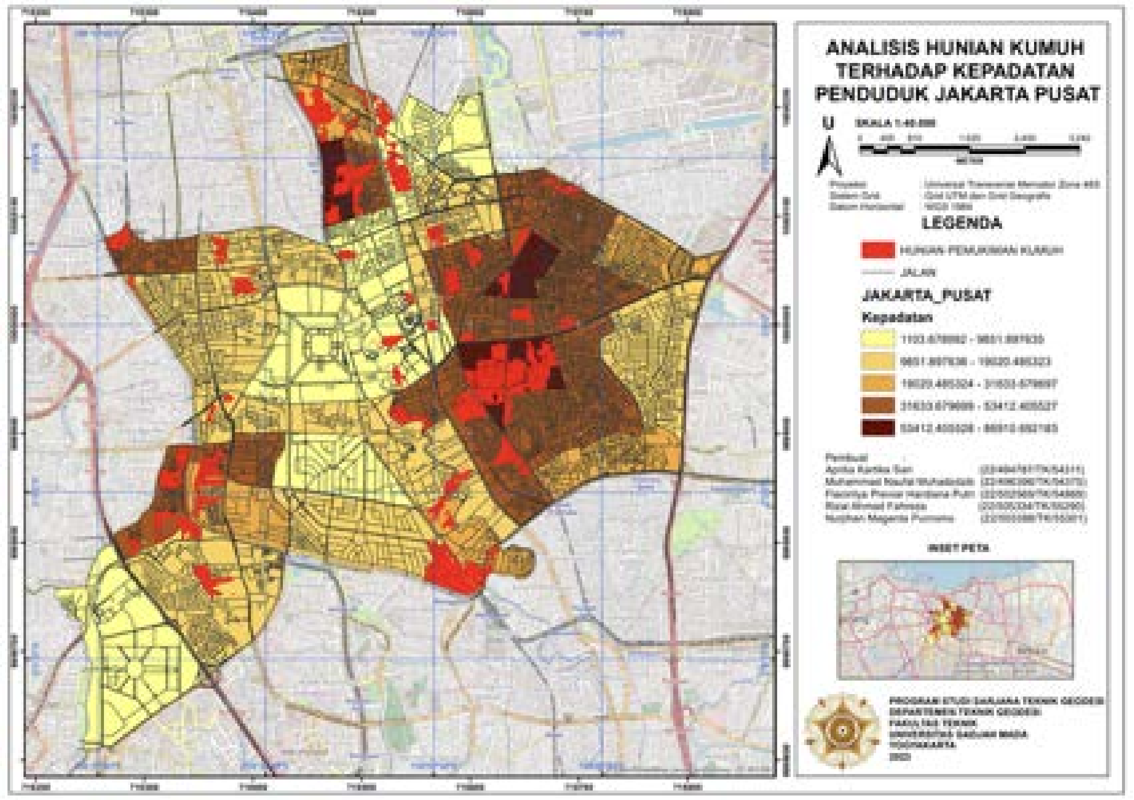
Data acquisition using GNSS RTK to capture topographic details. Processed 741 spatial points utilizing QGIS to digitize infrastructure features (buildings, roads, drainage) and generated contours TIN Mesh for a comprehensive 1:500 scale map.



ACADEMIC **PROJECTS**

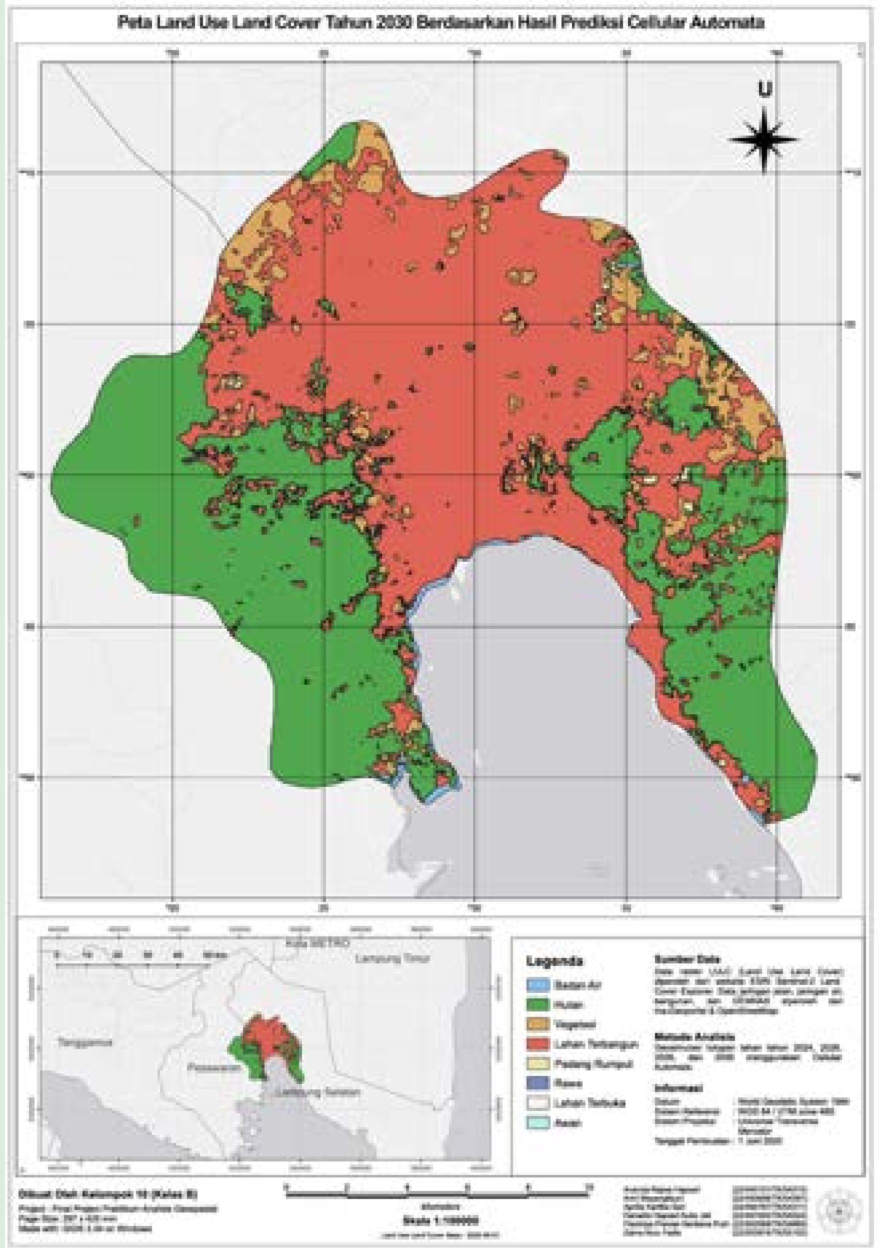
GIS (Geospasial Informartion System) & Geospatial Analysis

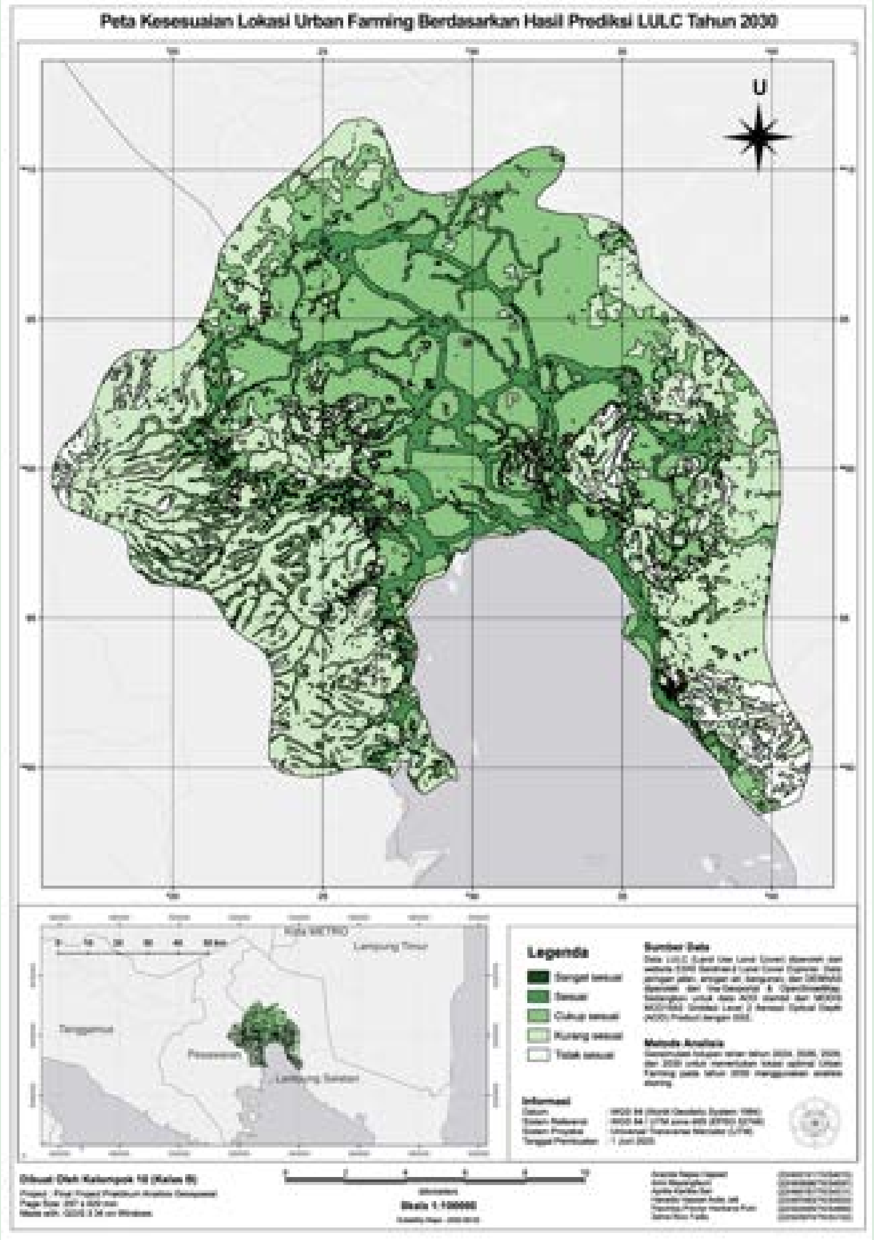




Urban Vulnerability & Demographic Analysis – Central Jakarta

Mapped the spatial distribution of informal settlements across multiple sub-districts in Central Jakarta. Overlaid these settlement footprints with demographic data to analyze the spatial correlation between high population density and urban vulnerability.

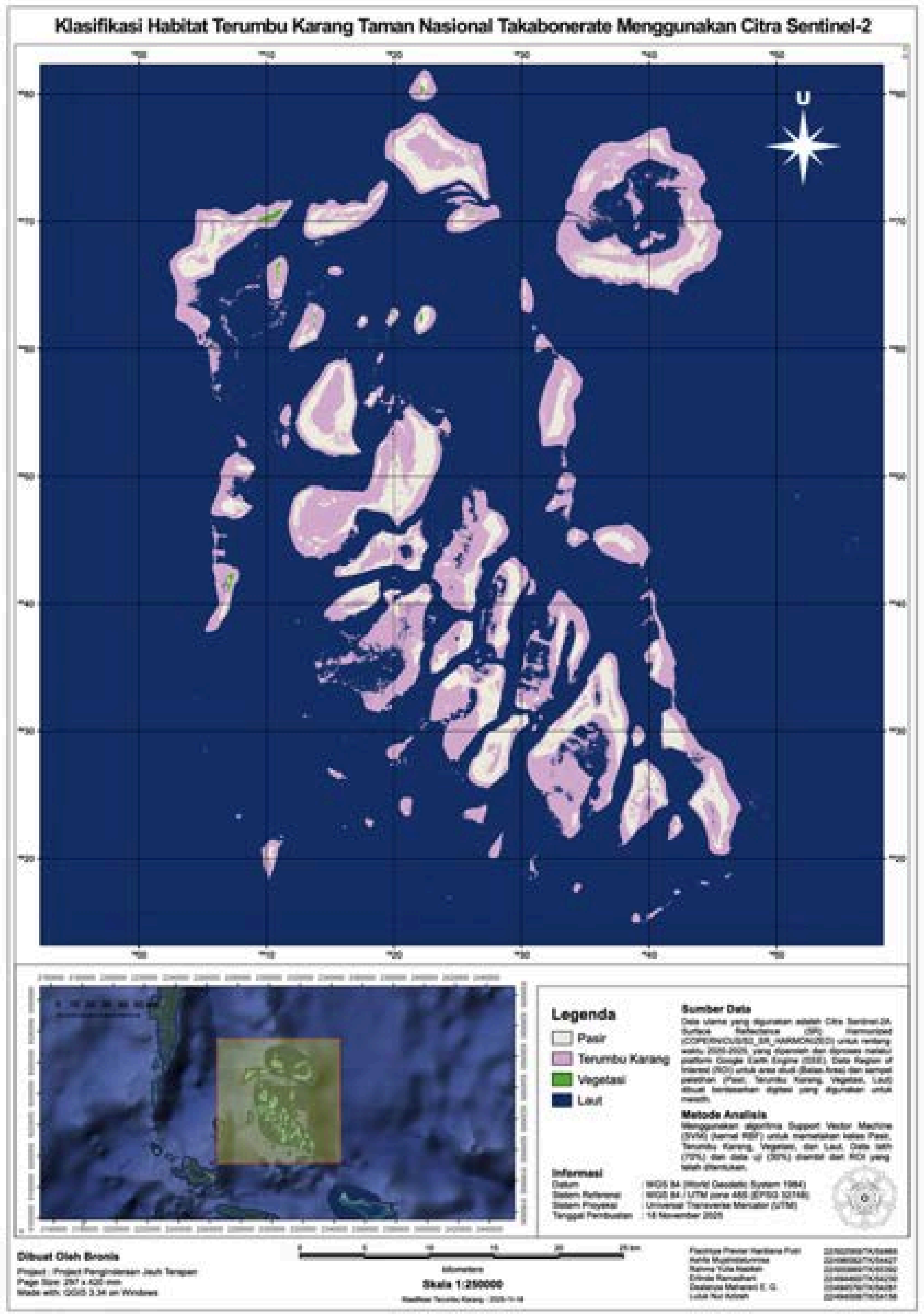




Simulated 2030 **Land Use and Land Cover** (LULC) configurations using **Cellular Automata** algorithms. Applied geospatial scoring and Google Earth Engine environmental data to this forecast to accurately delineate optimal **urban farming zones.**

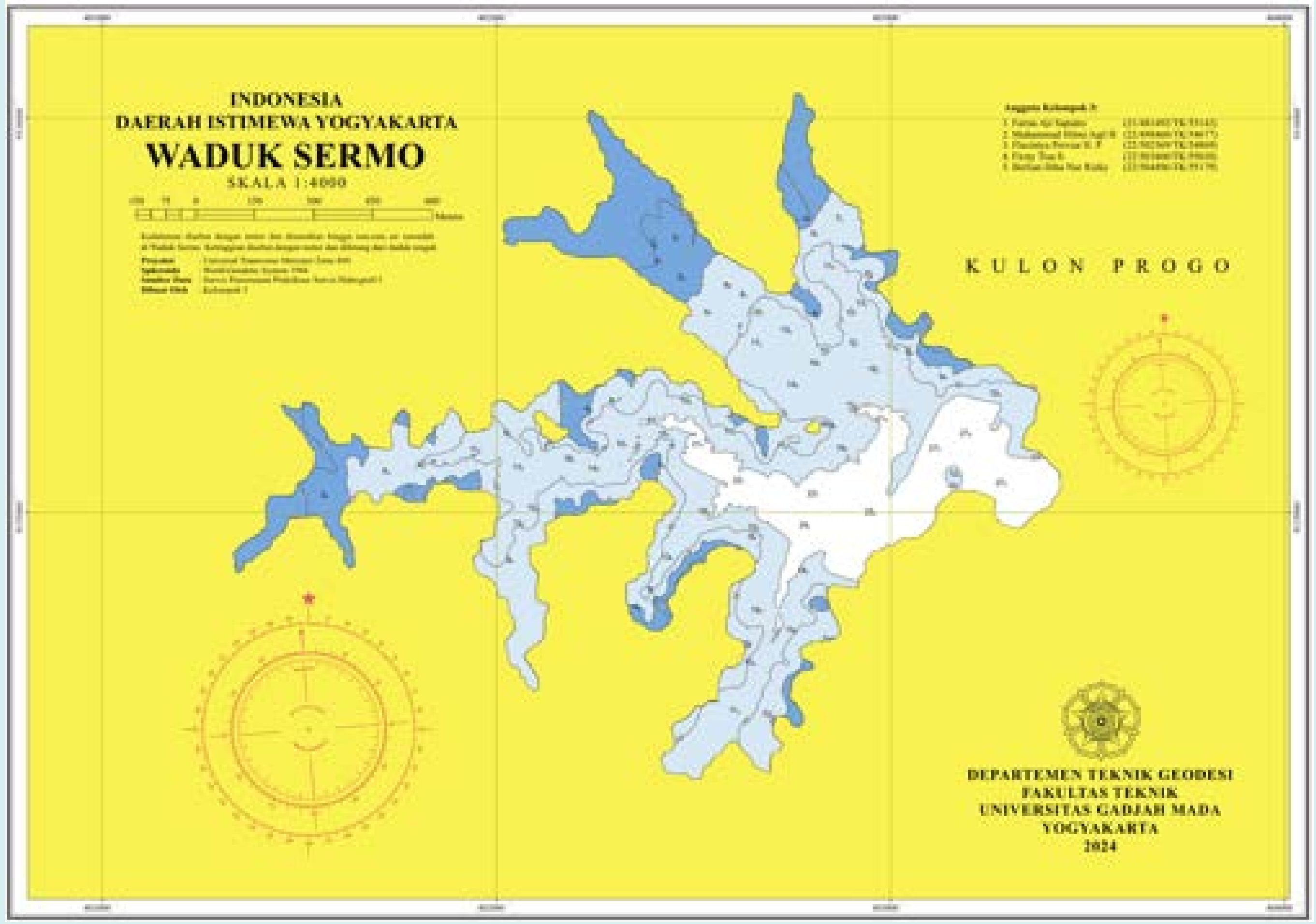
ACADEMIC **PROJECTS**

Remote Sensing & Hydrography



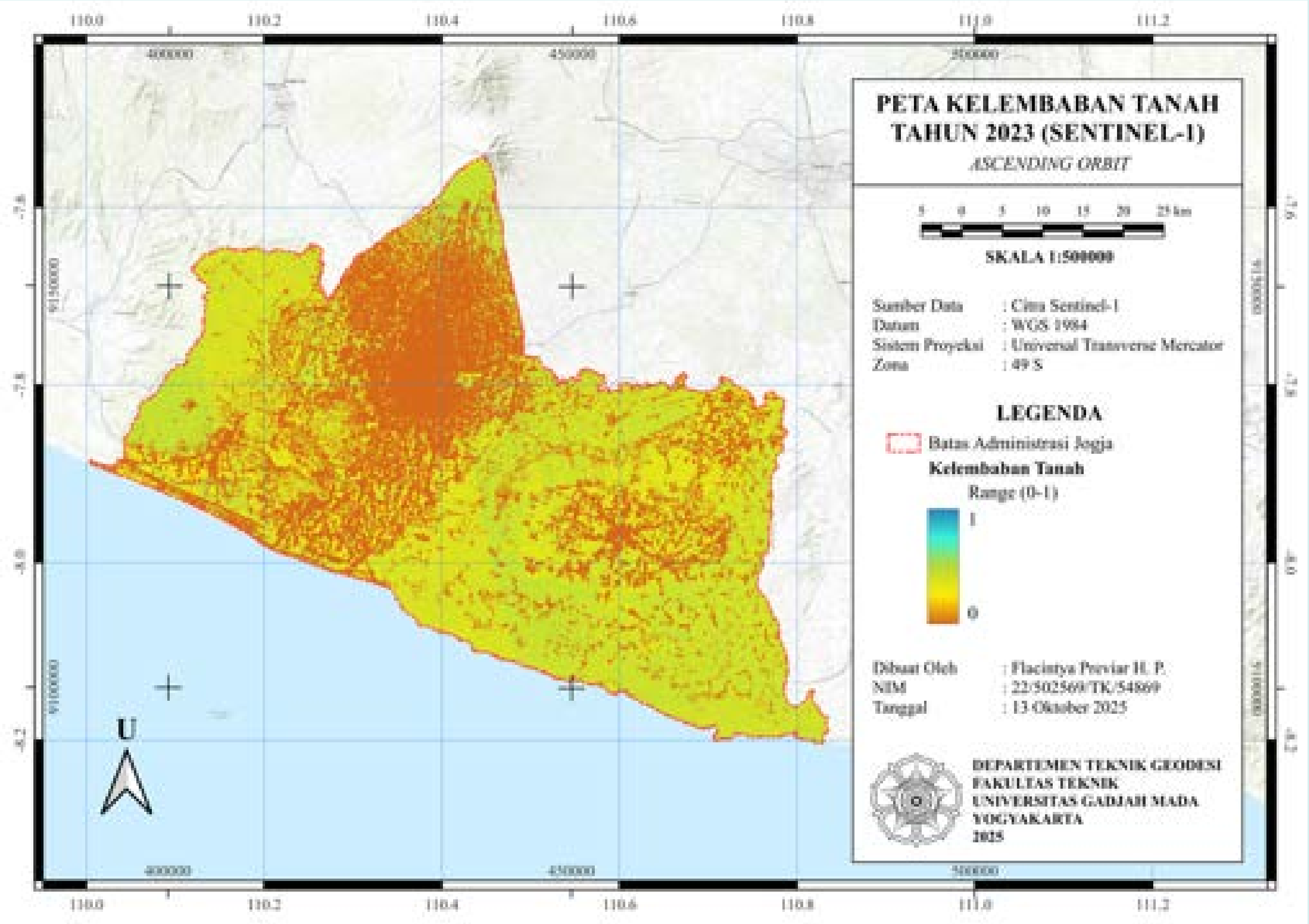
Coral Reef Habitat Classification – Takabonerate

Executed supervised classification of coral reef habitats in Takabonerate National Park by applying Support Vector Machine (SVM) algorithms via Google Earth Engine to Sentinel-2A imagery.



Bathymetric Map of Sermo

Generated a 1:4,000 scale bathymetric map of the Sermo Reservoir by conducting comprehensive depth sounding surveys to support advanced hydrographic analysis.



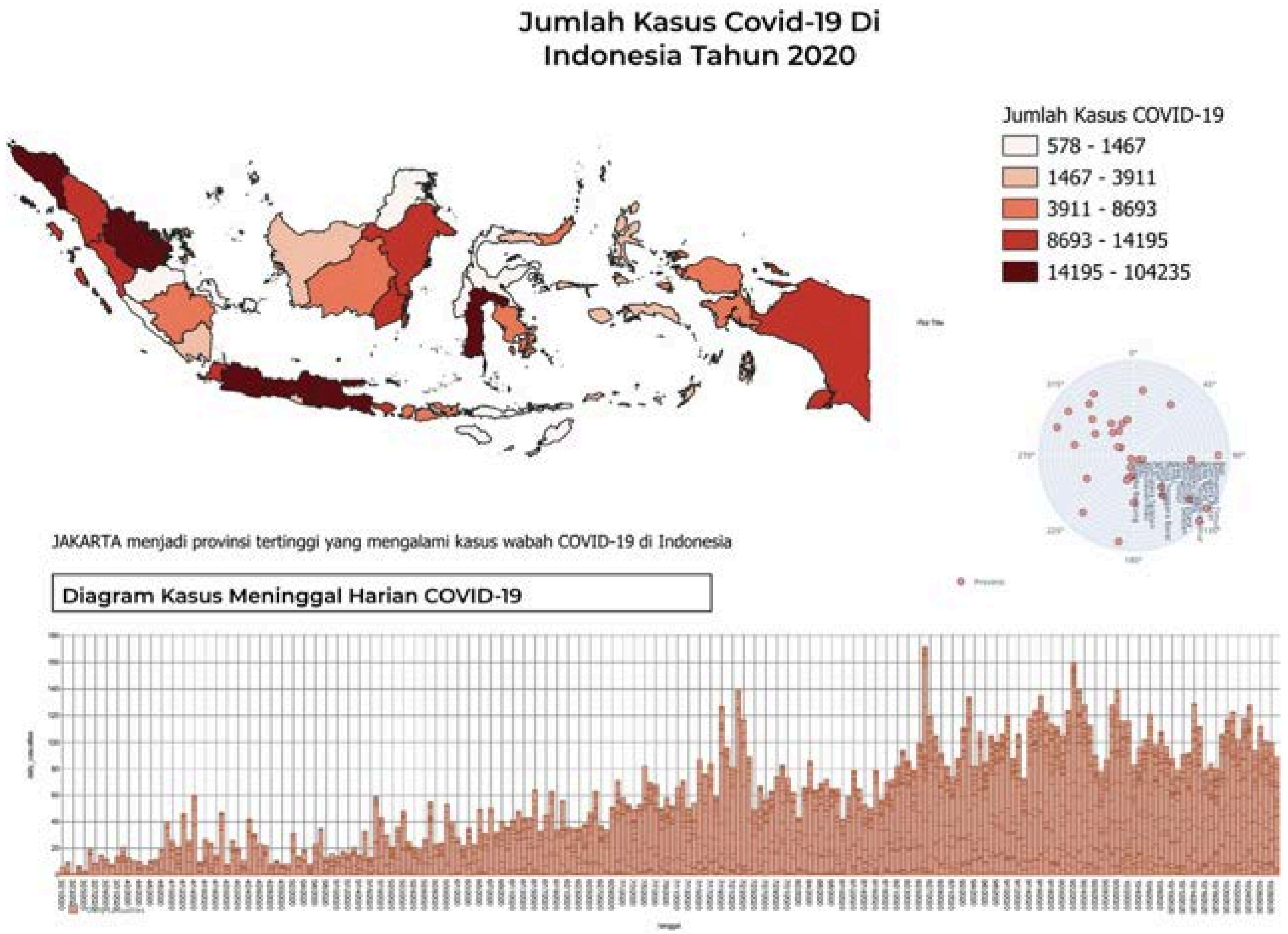
Sentinel-1 Soil Moisture Mapping

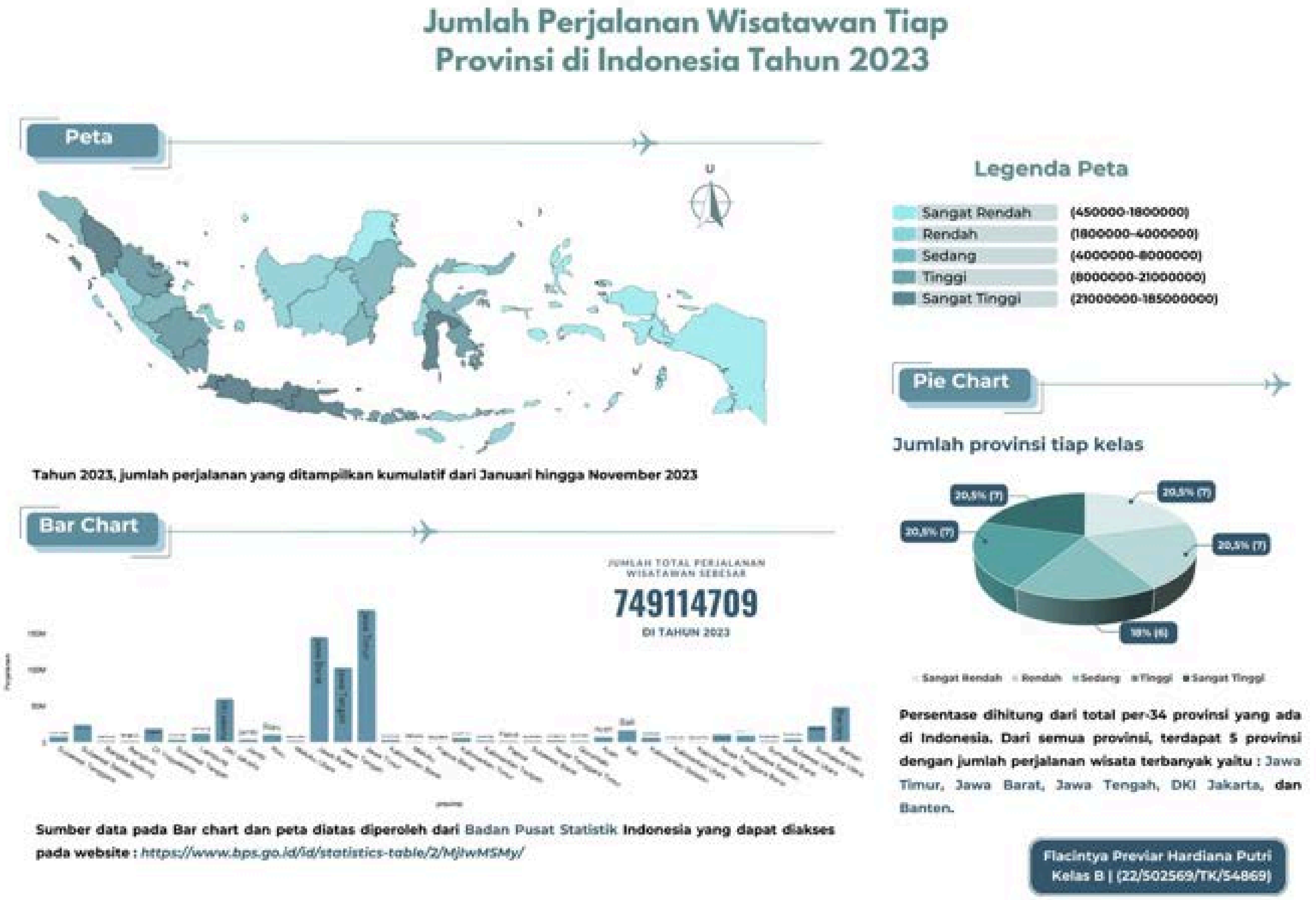
Mapped the spatial distribution of soil moisture across the Yogyakarta region by processing Sentinel-1 Synthetic Aperture Radar (SAR) ascending orbit imagery.

ACADEMIC **PROJECTS**

Cartography (Infographic Maps)

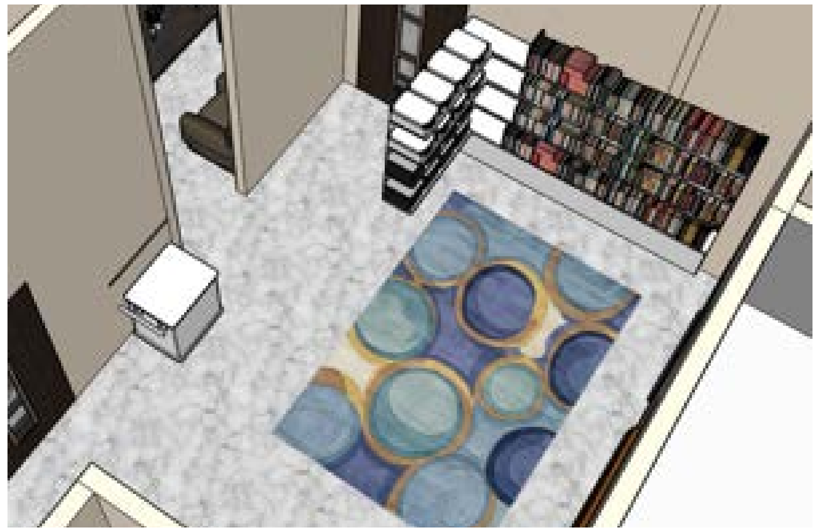
Designed a thematic infographic visualizing the spatial distribution and daily mortality trends of COVID-19 cases across Indonesian provinces using choropleth mapping techniques and a comprehensive geospatial dashboard illustrating domestic tourist travel frequencies across Indonesia by integrating thematic maps with statistical charts for demographic analysis.

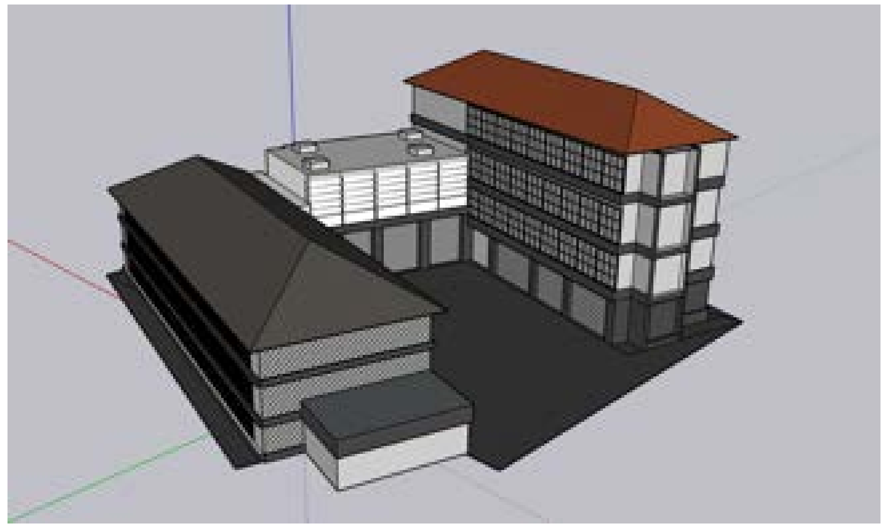












ACADEMIC **PROJECTS**

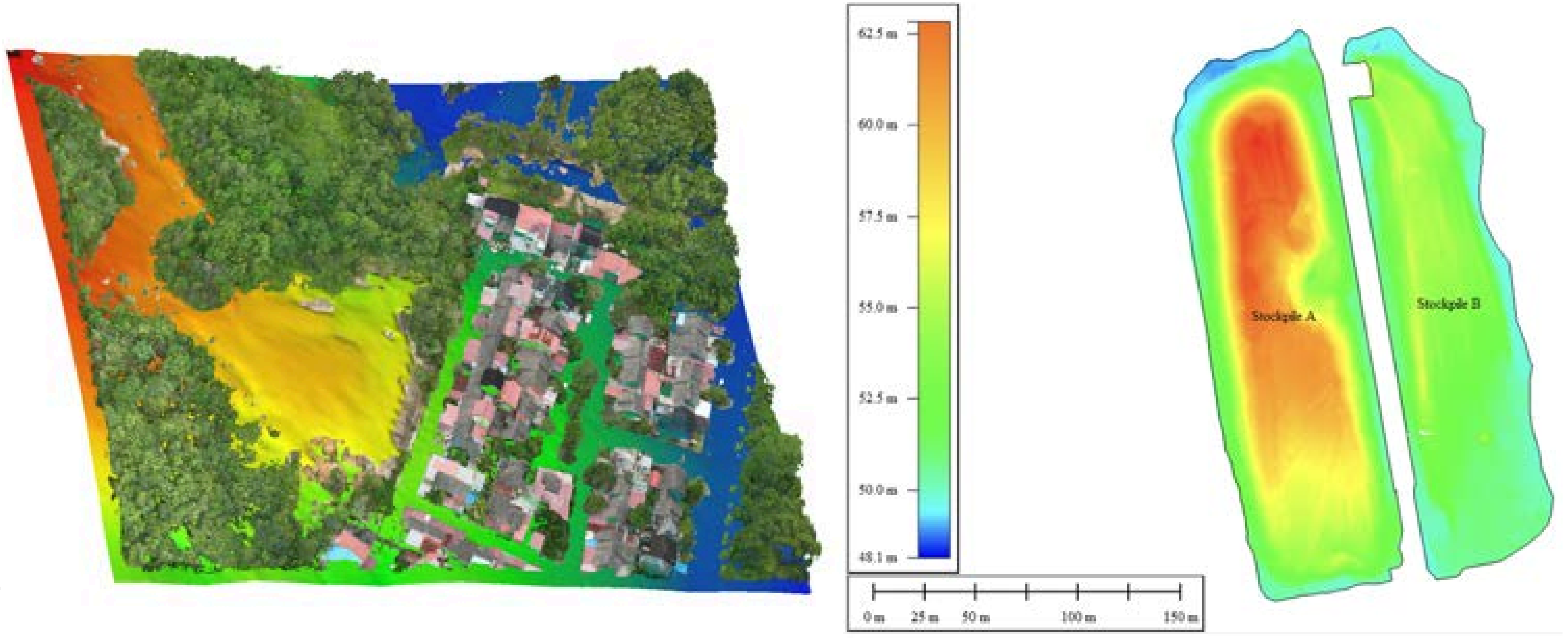
Creating 3D Building

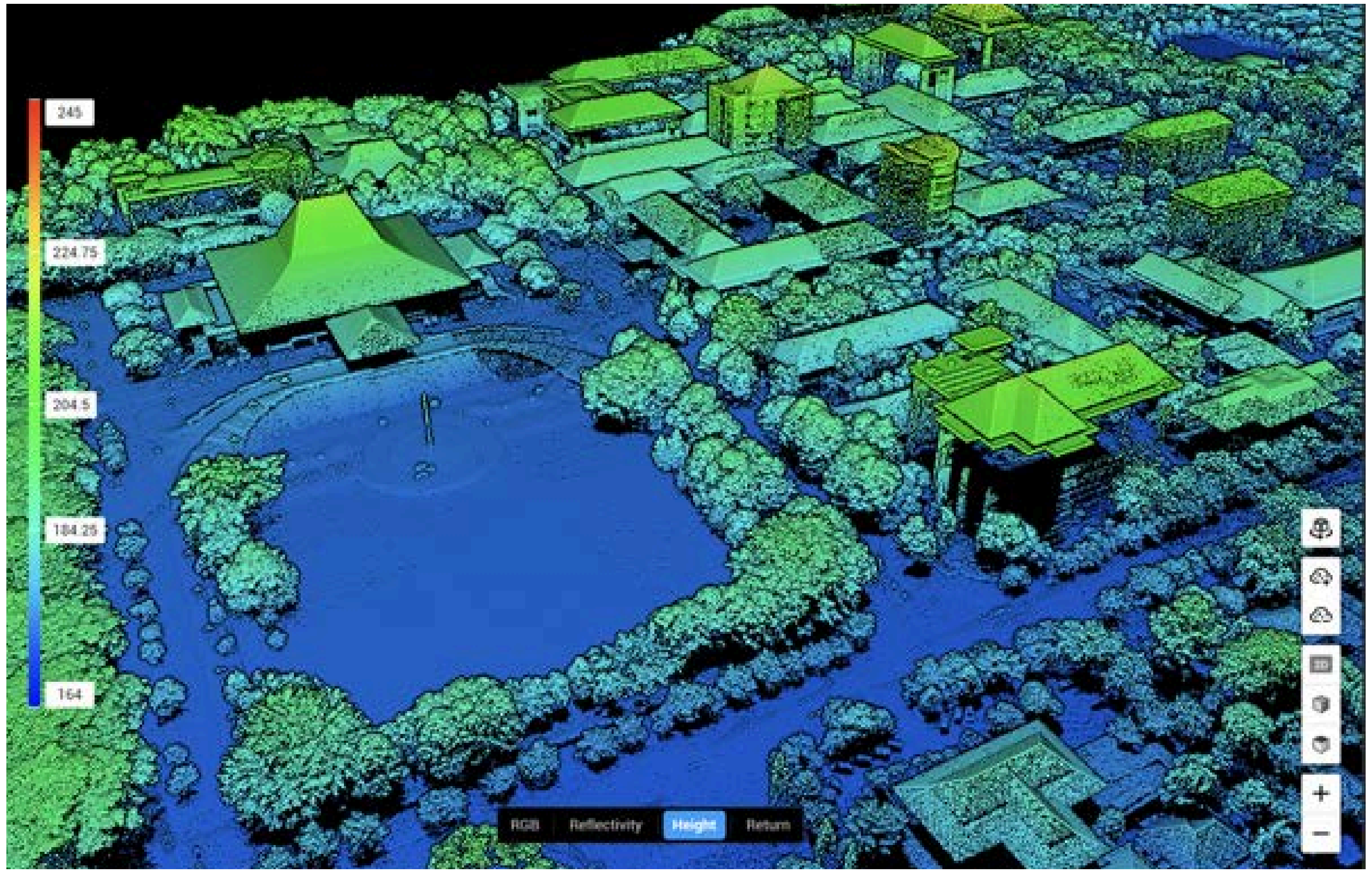
Designed and visualized a diverse range of 3D architectural models, including residential housing, high rise buildings, and interior spaces, utilizing SketchUp to translate 2D concepts into detailed, textured spatial representations.

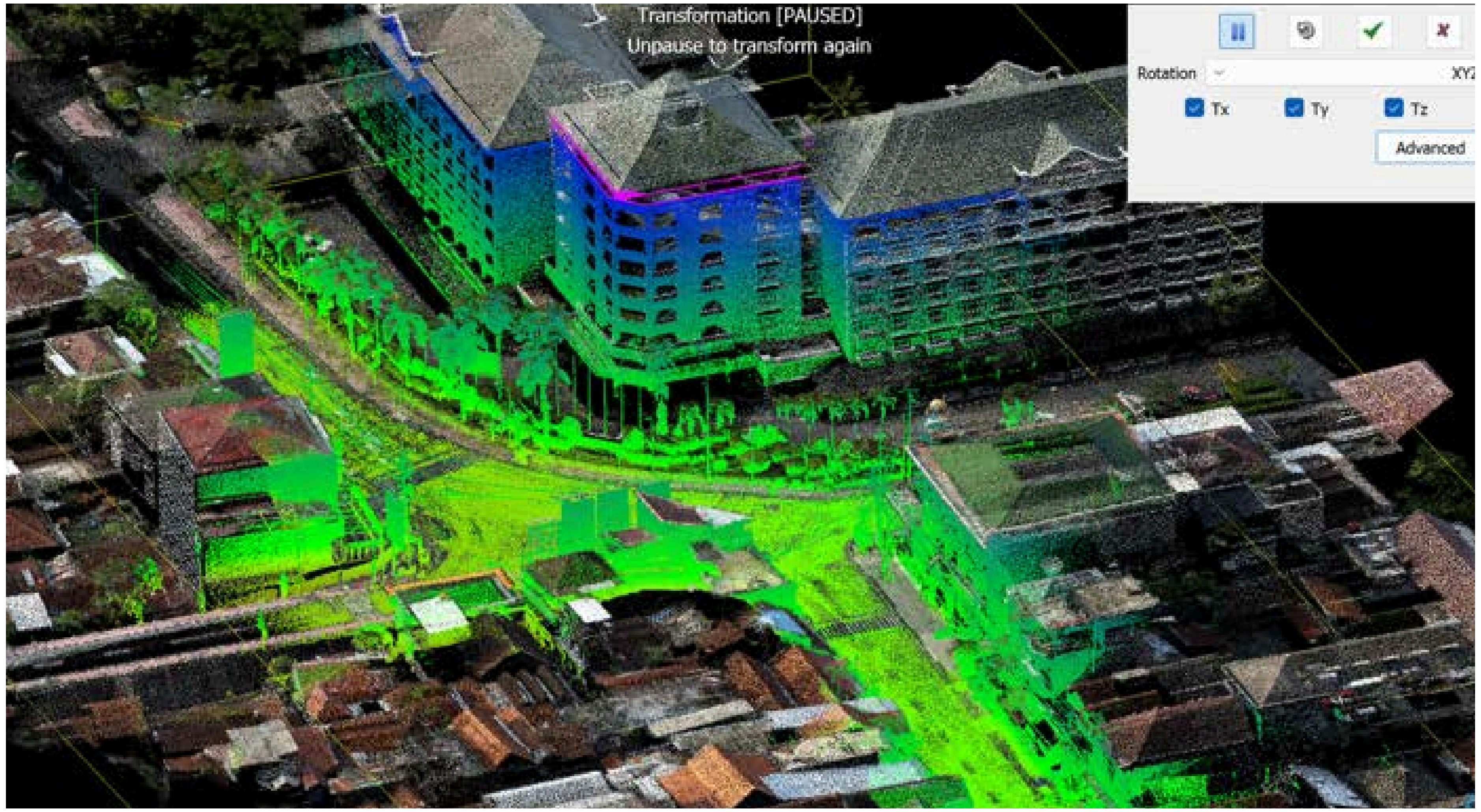
ACADEMIC **PROJECTS**

Lidar & SLAM data Processing

Processed aerial photogrammetry and LiDAR data to generate high-density point clouds and precise Digital Terrain Models (DTM). Utilizing Agisoft Metashape, CloudCompare, and Global Mapper, the project executed multi-sensor spatial data integration and extracted actionable DTMs to conduct accurate stockpile volumetric analysis.

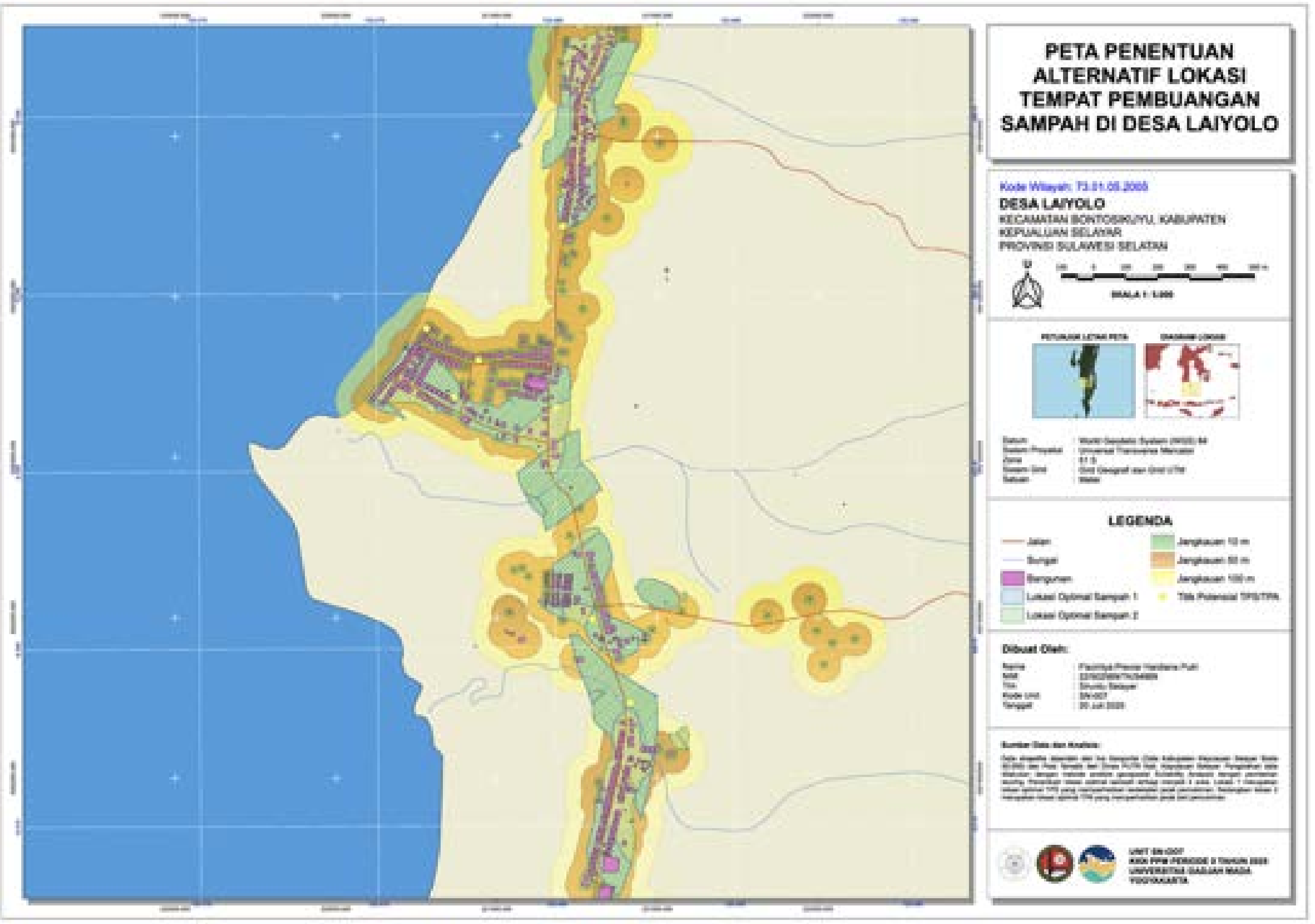




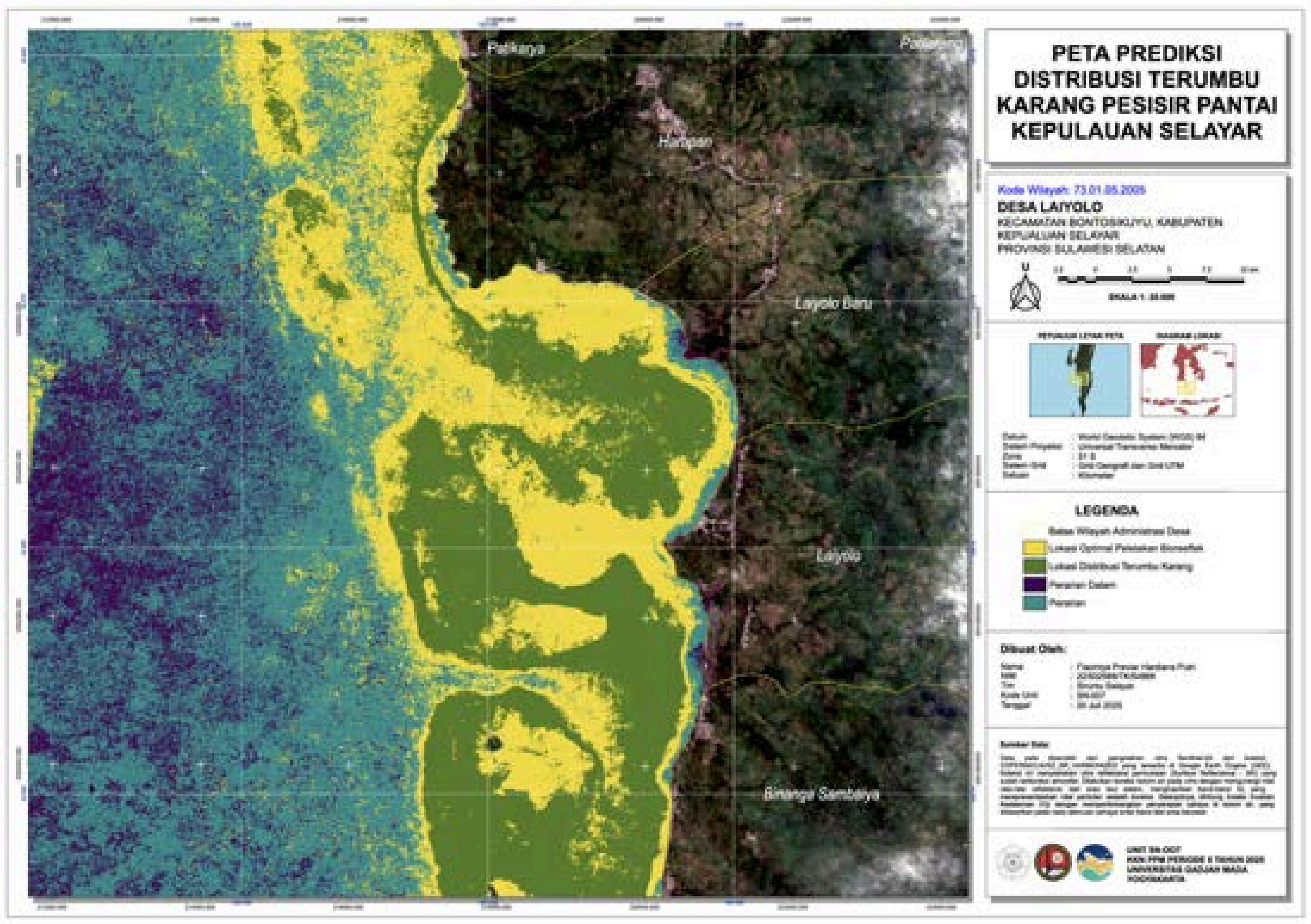


KKN-PPM PROJECT

Desa Laiyolo, Kec Bontosikuyu
Kab. Kepulauan Selayar



Mapped optimal waste disposal zones using geospatial suitability analysis. The project identifies two strategic locations based on residential proximity: a temporary transit point (**TPS**) and a final disposal site (**TPA**)



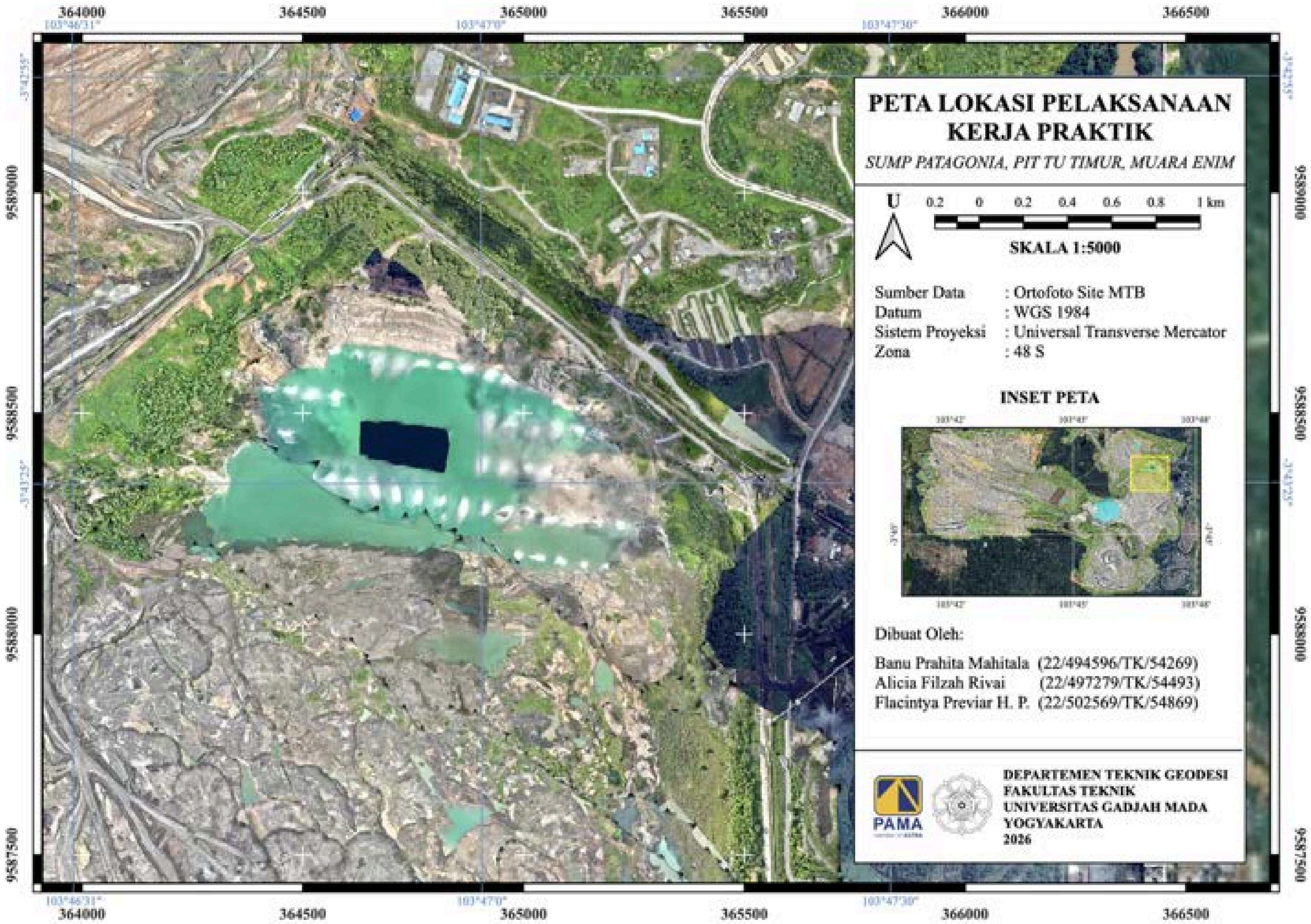
Mapped the predicted distribution of coastal **coral reefs** using **Sentinel-2A surface reflectance** imagery processed in GEE. Applied advanced geospatial techniques, including **water column correction** and **Depth Invariant Index** calculations, to accurately isolate and model underwater habitats.

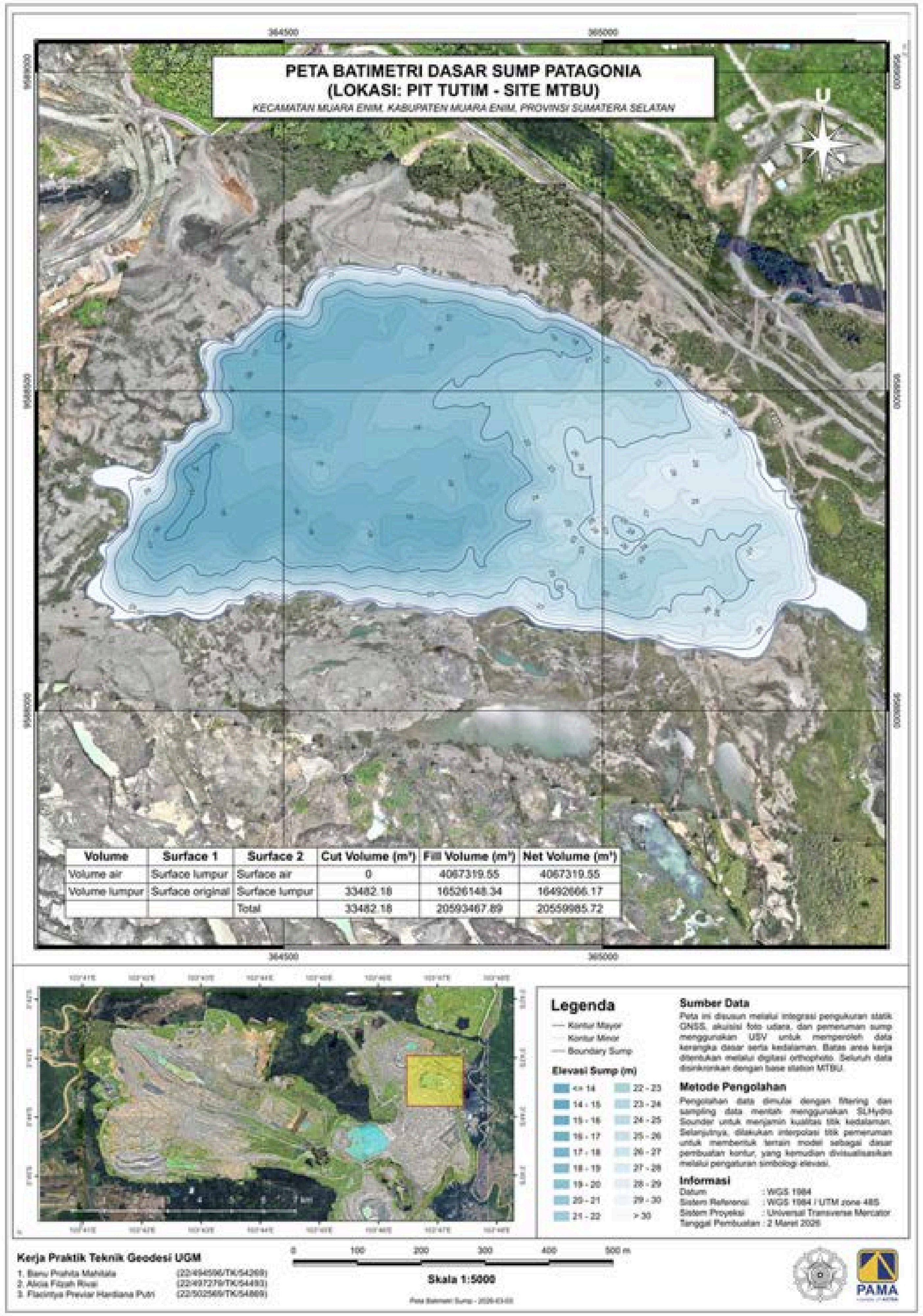
Developed spatial recommendations for water and drainage networks to mitigate water scarcity during seasonal transitions. Conducted advanced hydrological analysis using **Digital Elevation Models (DEM) and SAGA plugins** to accurately determine **flow direction, accumulation**, and **drainage patterns.**

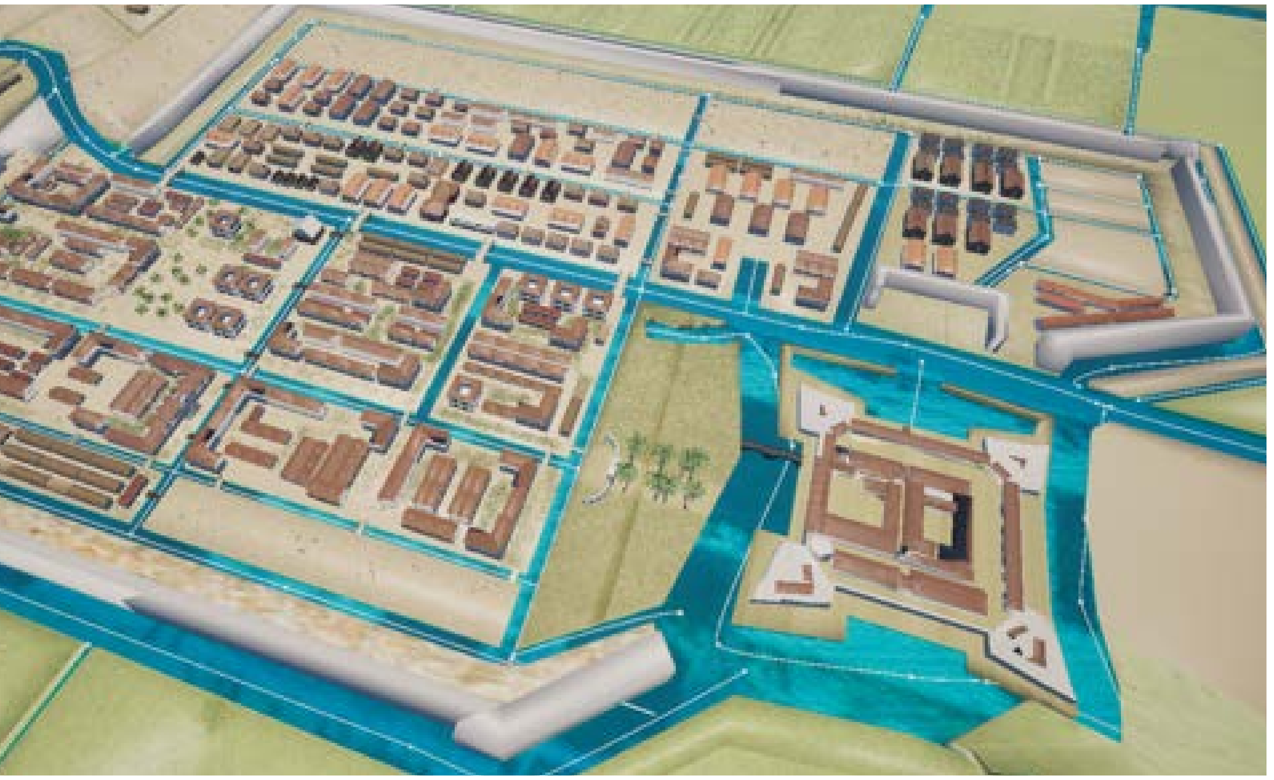
MINING**INTERNSHIP**

Pama Persada Nusantara – Site MTBU

- Executed advanced underwater terrain mapping by integrating static GNSS, aerial photogrammetry, and Unmanned Surface Vehicle (USV) depth sounding methodologies.
- Processed spatial data via SLHydro Sounder to generate a detailed bathymetric model, accurately calculating net volumes for water (4,067,319 m^3) and mud (16,492,666 m^3) to optimize mine water management.










3D MODEL **PROJECTS**

Heritage Reconstruction – LoD 3 – PBR Textures – Immersive Visual

Spearheaded the spatial reconstruction of **Old Batavia** by accurately georeferencing 17th-19th century colonial VOC maps against modern orthophotos. Engineered highly detailed, historically accurate **Level of Detail (LOD 3)** architectural models utilizing **SketchUp** and **Blender**, followed by the application of Physically Based Rendering (PBR) textures via **Adobe Substance 3D Painter** for photorealistic authenticity. Culminated the project by integrating all geospatial and 3D assets into **Unreal Engine 5,** developing a real-time, interactive platform.

3D MODEL **PROJECTS**

3D Modelling ‘Museum R.A Kartini Jepara’ LoD 4

Executed the **Level of Detail 4 (LOD 4)** 3D reconstruction of the **Jepara Museum**, focusing on the highly precise digitization of **interior** spaces and supporting architectural elements. Engineered high-fidelity 3D assets of intricate historical artifacts, vintage furniture, and virtual gallery spaces. By meticulously modeling structural interior features like central pillars, arches, and main doors, the project successfully transformed complex heritage environments into an immersive and spatially accurate digital experience.

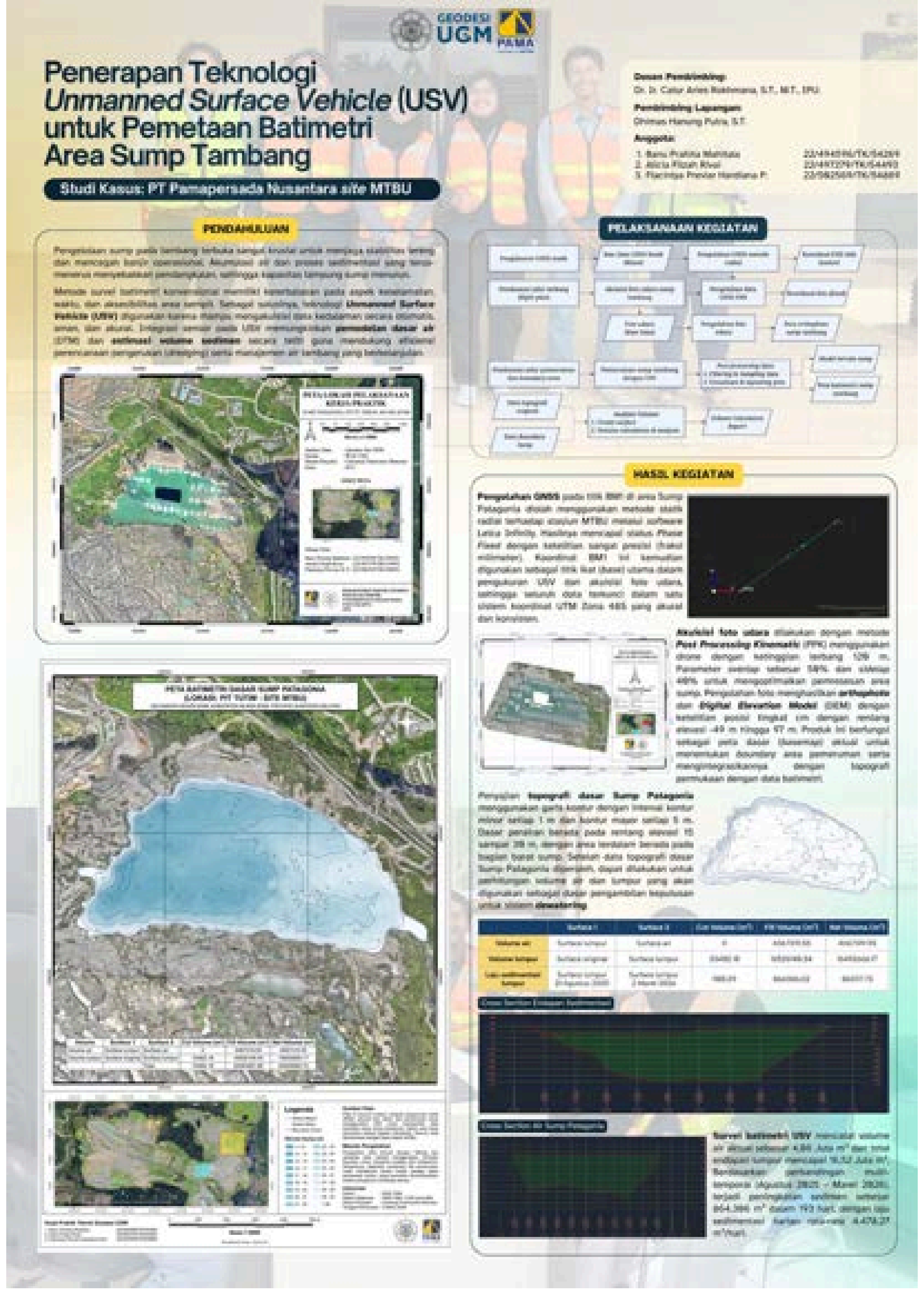
DIGITAL **ILLUSTRATION**

Event Poster Design

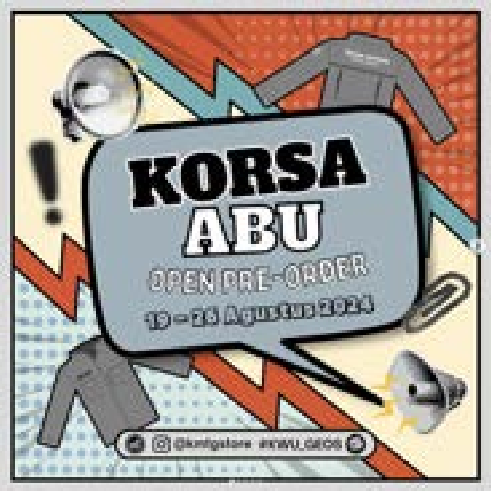
Designed various event posters with distinct themes, ranging from cultural festivals to modern art competitions. Focused on engaging digital illustrations and clear layouts to communicate information effectively.

POSTER **DESIGN**

Designed structured academic posters to summarize and present complex technical projects, focusing on clean layouts and clear data visualization.



SOCIAL **MEDIA**



Instagram @kmtgstore

Served as the Designer and Person in Charge (PIC) of Creative Content for the KMTG Entrepreneurship Department. Managed and designed engaging social media assets, including product promotions, event recaps, and carousel feeds for Instagram

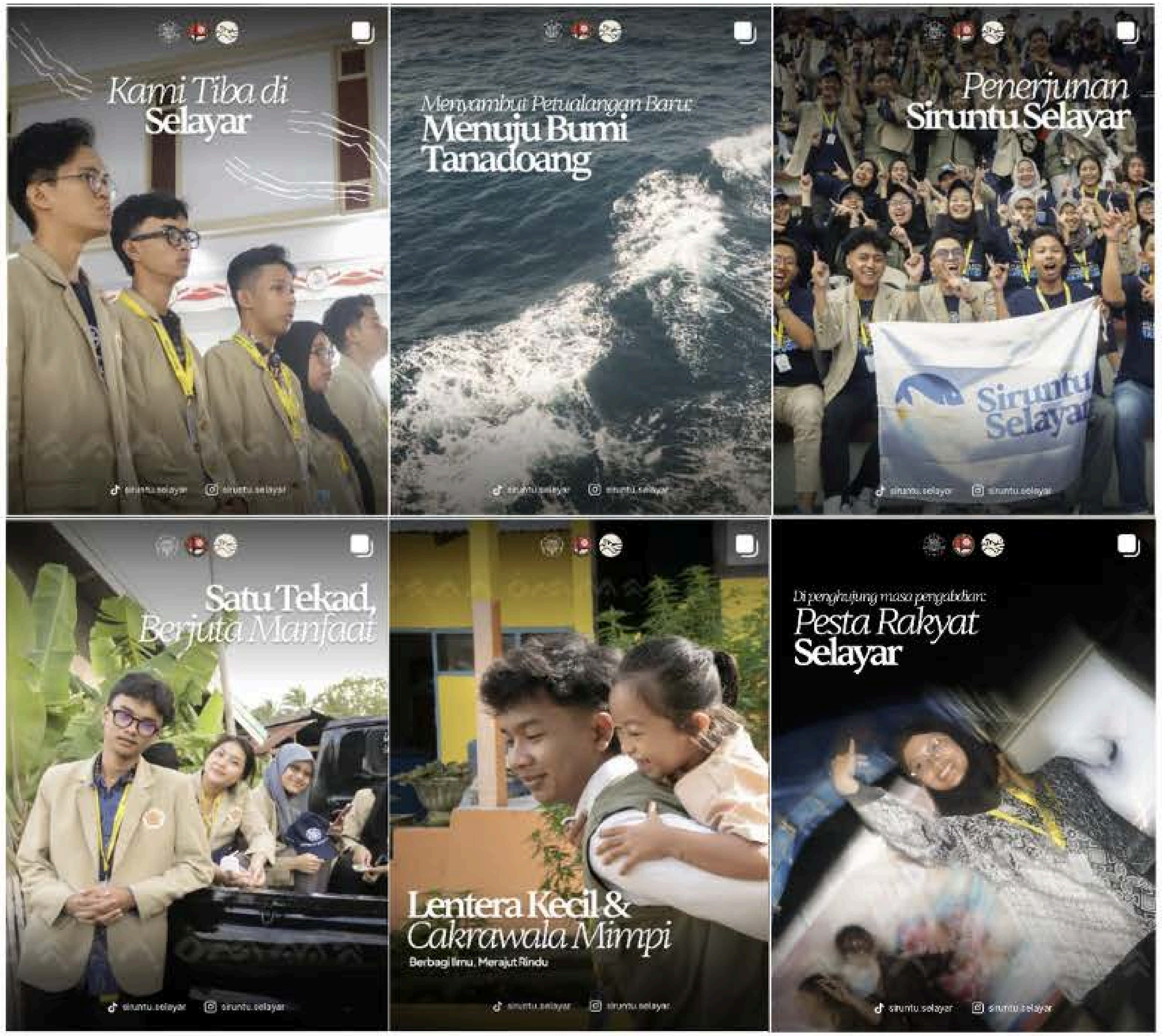
Instagram @kmtgstore

Instagram @kmtgugm

SOCIAL **MEDIA**

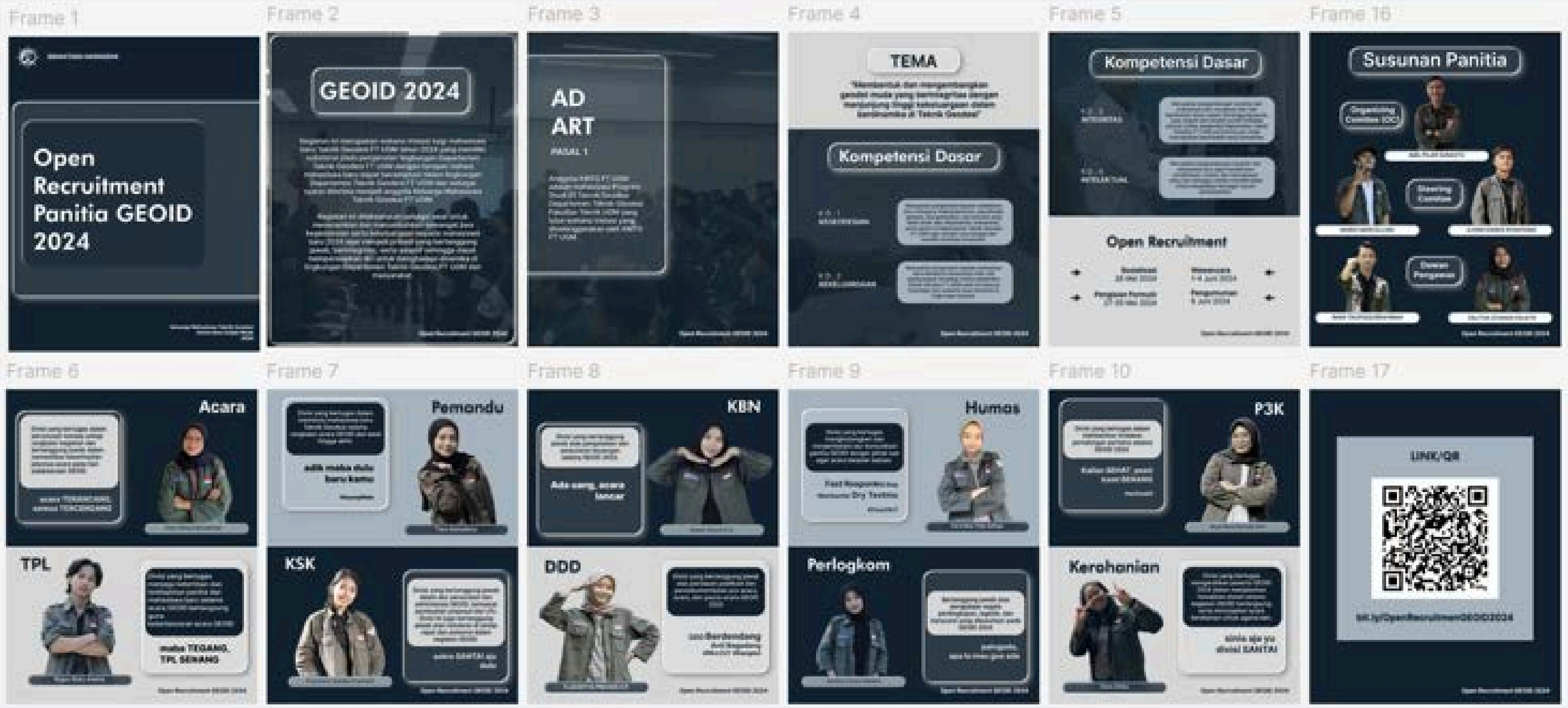
Instagram @siruntu.selayar

As the Head of Design and Documentation for KKN Siruntu Selayar, I directed the visual branding and managed the Instagram content, combining high-quality photography with cohesive layout designs.



BOOKLET / **GUIDEBOOK**

Create booklet and guidebook designs using figma & canva



PRODUCT DESIGN

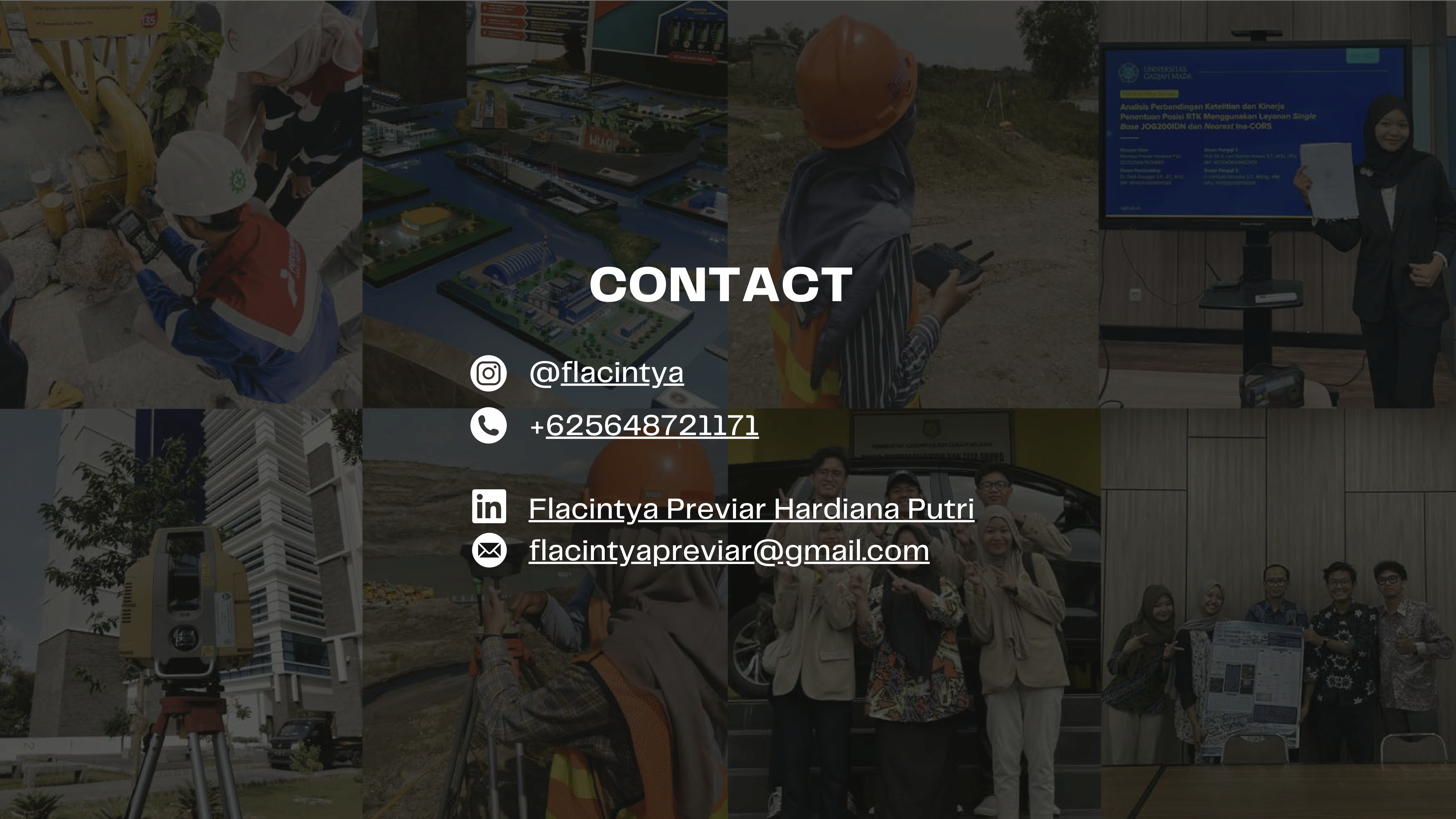
@kmtgstore

Designed a cohesive line of custom merchandise and professional apparel, including patterned lanyards, thematic vector stickers, and functional field uniforms (vests and shirts).

Lanyard Design

Sticker Merch Design

Shirt & Vest Design



CONTACT

@flacintya

+625648721171

Flacintya Previar Hardiana Putri

flacintyapreviar@gmail.com